

Endocrinology

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Sunday November 30, 2025



Outline (Lecture)

1. Hypercalcemia
2. MEN Syndromes
3. Hyperthyroidism
4. Hypothyroidism
5. Thyroid Disease in Pregnancy
6. Osteoporosis & Metabolic Bone Disease – **CMAJ Guidelines 2023**
7. Diabetes
8. Adrenal & Pituitary
9. Lipid Guidelines
10. Care for Transgender Patients
11. Reproductive Endocrinology
12. Thyroid Nodules & Cancer
13. Obesity Guidelines

Endocrinology 101

Before we start... Interpreting Labs:

***Normal or inappropriately normal?**

Always think critically about what the pattern of hormones should be,
not only whether a value is in the normal range

Example: A patient 4 weeks postop transsphenoidal craniopharyngioma patient presents weakness and fatigue. On exam has proximal weakness, hyporeflexia with delayed relaxation phase, pretibial myxedema, bradycardia and 10 kg weight gain.

TSH = 1.0

This patient likely has hypopituitarism after transsphenoidal surgery for a suprasellar mass like a craniopharyngioma. TSH is inappropriately normal. Free T3, T4 will be low if checked.

1. Hypercalcemia

Resources:

2022 Hyperparathyroidism Guidelines from the Fifth International Workshop

<https://asbmr.onlinelibrary.wiley.com/doi/epdf/10.1002/jbmr.4677>

Primary hyperparathyroidism: review and recommendations on evaluation, diagnosis, and management. A Canadian and international consensus: <https://pubmed.ncbi.nlm.nih.gov/27613721/>

MCQ 1- 2026

A 32-year-old woman is found to have mild hypercalcemia on routine testing. She is asymptomatic. On lab review her calcium was elevated 2 years ago also. She is on oral contraceptive pills only, no supplements. Which of the following is the most appropriate management?

- A.** Initiate cinacalcet therapy to lower calcium levels
- B.** Refer for parathyroidectomy to prevent complications of hypercalcemia
- C.** Advise no specific treatment and reassure the patient
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- E.** Recommend a low-calcium diet and vitamin D restriction

Lab Values	Reference Range
Calcium = 2.70	2.1 -2.6 mmol/L
Phosphate = 0.69	0.70 - 1.5 mmol/L
Cr = 60	53 - 97 μ mol/L
25OHD = 74	75 – 200 nmol/L
PTH = 6.7	1.6-6.9 pmol/L
Albumin = 40	35 - 50 g/L

Calcium Physiology: Pearl 1 & 2

1. In a *parathyroid*-mediated process, **serum calcium** and **phosphate** go in the opposite direction

e.g.

Hyperparathyroidism: $\uparrow\text{Ca}$ $\downarrow\text{PO}_4$

Hypoparathyroidism: $\downarrow\text{Ca}$ $\uparrow\text{PO}_4$

2. In a *vitamin D* mediated process, **serum calcium** and **phosphate** go in the same direction

e.g.

Osteomalacia: $\downarrow\text{Ca}$, $\downarrow\text{PO}_4$

Vitamin D *Excess: $\uparrow\text{Ca}$, $\uparrow\text{PO}_4$

(*ex. 1,25-hydroxyvitamin D excess from granulomatous disease)

Calcium Physiology: Pearl 3 & 4

3. When the calcium, phosphate and PTH **are all high**, think **kidney** (reduced ability to excrete phosphate)

e.g.

Tertiary Hyperparathyroidism (in long-standing renal failure):

↑Ca, ↑PO₄

4. Magnesium deficiency **reduces PTH secretion** and causes **PTH resistance**. Therefore think of hypomagnesemia in someone with apparent **hypoparathyroidism (or inappropriately normal PTH)**.

BONUS
Read on own

Lab Patterns in Calcium Problems

Disorder	Serum Calcium	Serum PO4	PTH	Urine Calcium
Primary HyperPTH	High	Low	High	High
Tertiary HyperPTH	High	High	High	High
Pseudohypoparathyroidism	Low	High	High	Variable
FHH	High	Low	High	Low (<0.01)
Lithium	High	Low	High	Low
Thiazides	High	N/Low	Low/High* * May unmask an underlying primary hyperparathyroidism	Low
Vitamin D excess	High	High	Low	High
PTHrP	High	Low	Low	High
Osteolysis	High	High	Low	High
Vitamin D Deficiency	Low	Low	High	Normal / High

1° Hyperparathyroidism (PHPT): Surgical Management

Know when to send someone for surgery...

Symptomatic PHPT = Surgery!

Asymptomatic PHPT= “Stay The Fudge Away U Stupid Calcium”

AT DIAGNOSIS:

Serum total calcium > 0.25 mmol/L above upper limit

T-score ≤ -2.5 at L-spine, total hip, femoral neck or ***distal 1/3 radius***

Fractures (Vertebral only; by X-ray, CT, MRI or VFA)

Age < 50

Urine calcium >6.25 mmol/d (>250mg/d) in women or >7.5 mmol/d in men (>300mg/d)

Stones or nephrocalcinosis by x-ray, ultrasound, or CT

Creatinine clearance < 60 mL/min (stage 3 CKD)

JBMR 2022

Always Rule Out FHH!

Familial Hypocalciuric Hypercalcemia

- **FHH** results from **inactivating mutations in the calcium-sensing receptor (CaSR)** gene, leading to a higher “set point” for calcium homeostasis.
- **MUST DO A URINE CALCIUM before surgical referral for hyperparathyroidism**

	Urine Calcium	Serum Calcium
FHH	LOW ↓	HIGH ↑
Hyperparathyroid	HIGH ↑	HIGH ↑

- **Urine calcium:creatinine ratio < 0.01 (less than 1%!) → think FHH; PHPT generally have UCa:Cr ratio >0.02**

$$\frac{\text{Urine Calcium (mmol/l)} \times [\text{Serum Creatinine (umol/l)} / 1000]}{\text{Serum Calcium (mmol)} \times \text{Urine Creatinine (mmol/l)}}$$

1° Hyperparathyroidism (PHPT): Medical Management

If patient **NOT** a candidate for surgery (e.g. per ENT, or too frail for surgery) :

Medical Management*

- ! Correct vitamin D deficiency/insufficiency: target serum 25-OH vit D to >75 nmol/L
- Calcium intake should be consistent with nutritional guidelines (1000-1200 mg/d)
- **Bisphosphonates** and **denosumab** if indicated i.e. high risk per Osteoporosis Guidelines
 - These can INCREASE BMD in primary hyperparathyroidism
- **Cinacalcet** (\$) is effective in reducing serum Ca and should be considered for symptomatic PHPT if surgery is not an option. **May combine w BP or denosumab** in selected pts (to reduce Ca)
- *Bottom Line: Surgery is the only cure for PHPT. Never use medical mgmt as an alternative to surgery when patient is fit for OR. There is no fracture data (only BMD improvement) with the existing medical treatments.

2° & 3° Hyperparathyroidism

Secondary Hyperparathyroidism

- **Appropriate increase in PTH** due to hypocalcemia or **vitamin D deficiency**
- **PTH is appropriately working to absorb all the urine calcium/salvage calcium level**
 - **TIP #1** : Postop Gastric Surgery (ex. gastric bypass/bilroth/whipples surgery where a portion of stomach is removed)
 - You CANNOT use Calcium Carbonate as a supplement (there is no acidity to absorb this!)
= use Calcium Citrate
 - **TIP #2**: Pt w Renal disease (CKD)
 - treat with Vitamin D, phosphate restriction, non-calcium phosphate binders
 - Cinacalcet for 2° hyperthyroidism in CKD on dialysis (with target PTH levels depending on CKD stage; often in consultation with Nephrology) – cost \$\$ prohibitive in many patients

Tertiary Hyperparathyroidism

- Longstanding hypocalcemia (appropriate stimulus for PTH release) → parathyroid glands can become **autonomous, secreting PTH even when calcium corrected**
- Usually in the setting of **end-stage renal disease or post-transplant**

Indications for Surgery in Tertiary Hyperparathyroidism

- Refractory hyperPTH despite vit D analogues/ calcimimetics (No absolute #/PTH cutoff, KDIGO defines as PTH still rising, symptomatic)
- Hypercalcemia severe / symptomatic
- Calciphylaxis
- Progressive bone disease

Remarkably the KDIGO 2017 guidelines don't give a lot of guidance here, largely because there are no high quality RCTs showing difference in outcome of medical vs surgical therapy in this population.

Spotlight: Postop Hungry Bone Syndrome

Chronic high PTH demineralizes bone

- Severe cases include osteitis fibrosa (↑bone turnover, bone resorption, bone can fill with fibrous tissue / brown tumours)
- Sudden drop in PTH after parathyroidectomy in 2o, 3o HPTH causes acute mineralization of bone
= *Profound and sustained hypocalcemia, hypophosphatemia, hypomagnesemia within 1-4 days of surgery*
- Patients usually require ICU level care for high dose PO calcium, calcitriol, IV calcium infusions, other electrolyte replacement

4.2.5: In patients with CKD G3a–G5D with severe hyperparathyroidism (HPT) who fail to respond to medical or pharmacological therapy, we suggest parathyroidectomy (2B).

BONUS

Read on own

Hypoparathyroidism

Diagnosis: Hypocalcemia in the presence of undetectable, low or inappropriately normal PTH (2 occasions > 2 weeks apart)

Differential:

Acquired

- **Hypomagnesemia** → PTH resistance
- **Hypermagnesemia** → reduce PTH synthesis/secretion
- Post-surgical (common complication post-total thyroidectomy) or post-radiation
- Infiltrative disease (sarcoid, amyloid, cancer metastasis)
- Autoimmune polyglandular syndrome type 1 (APS-1) (Whitaker's Triad: chronic mucocutaneous candidiasis, Addison's disease, & hypoparathyroidism) - *AIRE* mutation

Congenital

- Pseudohypoparathyroidism (genetic mutation in *GNAS* gene)
- DiGeorge Syndrome / 22q11.2 deletion syndrome: parathyroid agenesis

Note: Hypocalcemia outside of hypoparathyroidism can also occur in Extravascular sequestration – Hyperphosphatemia, Pancreatitis, Osteoblastic metastases

Management:

- 1st line = Conventional therapy w/ oral calcium & active vitamin D (calcitriol or alfacalcidol)
- PTH therapy can be considered if conventional therapy insufficient

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C. She has FHH. Her PTH is inappropriately high normal. A 24h urine would confirm with low urinary calcium.

2. Multiple Endocrine Neoplasms (MEN)

MEN 1 (PPP)	PARATHYROID	PITUITARY ADENOMA	PANCREATIC (insulin, VIP, <u>gastrin</u>)
MEN 2A (PMP)	PARATHYROID	MEDULLARY THYROID CANCER	PHEOCHROMOCYTOMA
MEN 2B (MMP)	MARFANOID, MUCOSAL NEUROMAS	MEDULLARY THYROID CANCER	PHEOCHROMOCYTOMA

**** When you see one, think of the others (especially in a young patient)**
**** All Autosomal Dominant**

3. Hyperthyroidism/Thyrotoxicosis

Highest yield resources:

2016 ATA Thyrotoxicosis Guideline

<http://online.liebertpub.com/doi/full/10.1089/thy.2016.0229>

2017 ATA Thyroid Disease in Pregnancy Guideline

<https://www.liebertpub.com/doi/full/10.1089/thy.2016.0457>

MCQ 2 - 2026

A 29-year-old woman, 8 weeks pregnant, presents with palpitations, heat intolerance, and unintentional weight loss over the past month. On exam, she has diffuse goiter and mild exophthalmos. Labs show: TSH: <0.01 mIU/L, Free T4: 45 pmol/L (elevated), Positive TRAB

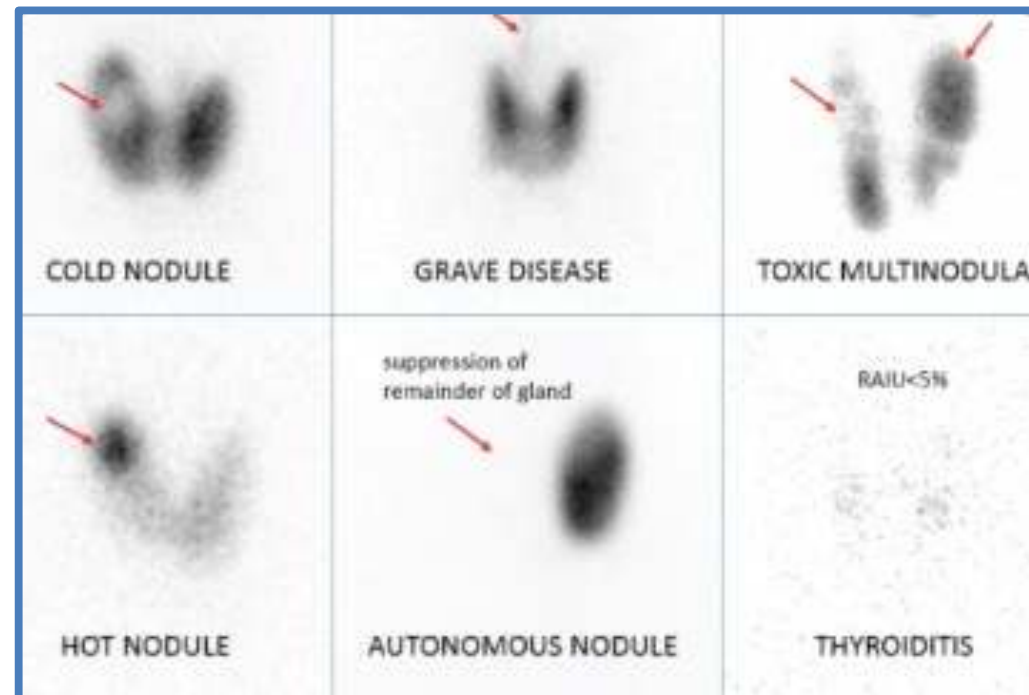
Which of the following is the preferred first-line antithyroid medication during the first trimester?

- a) Propranolol
- b) Methimazole
- c) Propylthiouracil (PTU)
- d) Radioactive iodine (I-131)

Hyperthyroidism* – Pearl # 1

- High radioactive iodine uptake (RAIU) (usually >25%) = increased **endogenous production of thyroid hormone** (e.g. Graves disease, toxic multinodular goiter)
- Low radioactive iodine uptake = extra thyroid hormone **without increased endogenous production** → *exogenous ingestion* or inflammatory leak (e.g. acute, sub-acute, post-partum, or amiodarone-induced thyroiditis)*

**RAIU may be falsely low if there is an interfering factor (e.g. recent iodine load via IV contrast or amiodarone, use of thionamide medications)*



Hyperthyroidism – Pearl # 2

- **Ophthalmopathy = Graves disease**
 - no matter what the RAIU result!
- If the RAIU is low *even with these clinical features*, **the test is usually unreliable because of:**
 - CT scan (iodinated contrast)
 - Amiodarone (iodine load)
 - Iodine intake (e.g. kelp)

Patient with thyrotoxicosis/
untreated hyperthyroidism
requiring CT scan with contrast?

Consider pre-treating with anti-
thyroid drugs prior to CT.

Hyperthyroidism – Pearl # 3

- Use beta-blockers (e.g. propranolol) for symptomatic patients (“especially elderly, those with resting HR > 90 or CVD”)
- It takes **4-6 weeks to see full effect of anti-thyroid drugs** (e.g. methimazole (MMZ) or propylthiouracil (PTU))

Hyperthyroidism – Pearl # 4

Treatment Options for Graves: thionamides, surgery (total thyroidectomy), or RAI ablation; latter two renders the patient hypothyroid → need levothyroxine replacement lifelong

Medical Management:

- Use **MMZ** instead of PTU because less hepatotoxic EXCEPT in the following situations:
 - First trimester of pregnancy (risk of aplasia cutis & cleft palate)
 - Thyroid storm
 - **Minor** MMZ reactions (if severe, then shouldn't use anti-thyroid drugs at all)

Hyperthyroidism – Pearl # 5

- Be aware of biotin supplementation, if patient is taking >30 µg of biotin a day it can interfere with thyroid assays in lab and cause falsely low or high results

Many patients taking biotin [vitamin B7] at doses of 5-10mg (5000-10000 µg) “for hair and nails”.

Many immunoassays in the lab use biotin-streptavidin binding systems, so high [biotin] interferes.

Other serum results affected:

False ↓ TSH, PTH, bHCG, Troponin, NT-pro-BNP

False ↑ free T3/T4

There are others – check with your lab! Advise pts to stop biotin at least 72h before bloodwork.

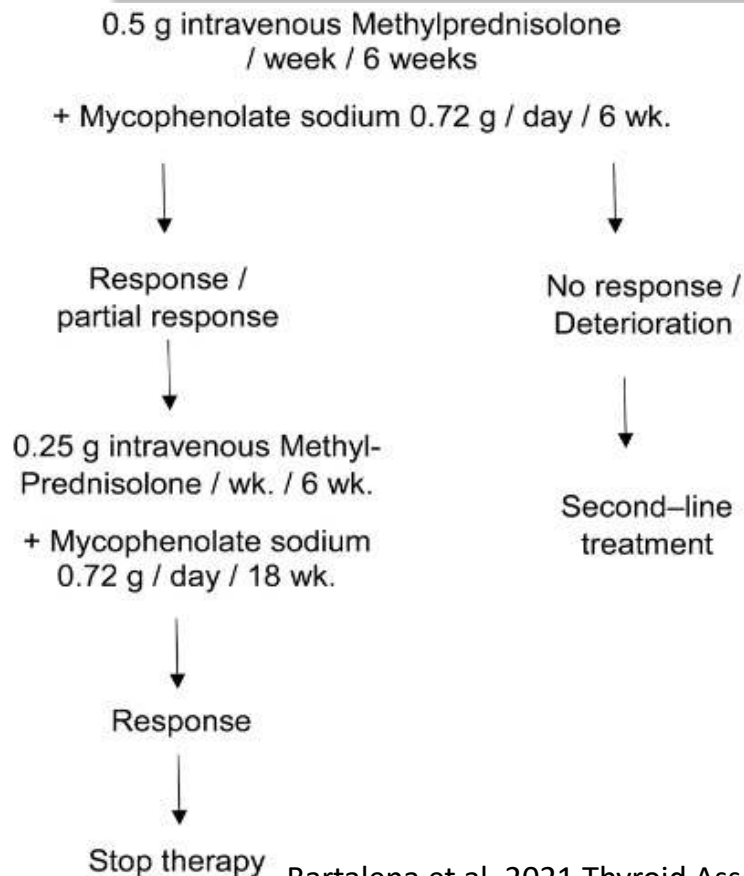
Counselling on Anti-Thyroid Drugs

Drug Reaction	MMZ	PTU	Action	Management Notes
Hepatotoxicity*	YES	YES (worse)	STOP & can't switch	Discontinue if transaminases >3x ULN * *some transaminitis often also a result of thyrotoxicosis before ATD, but typically not 3x ULN
Agranulocytosis (look out for fever/sore throat!)	YES (more likely dose related)	YES (not dose related)	STOP & can't switch	Granulocyte colony stimulating factor, steroids, antibiotics (if febrile) and/or supportive care
Vasculitis (polyarthrititis, purpuric skin lesions, pulmonary +/- renal)	YES	YES	STOP & can't switch	Rx - Glucocorticoids or other immunosuppressive therapy
Serious allergic rxn	YES	YES	STOP & can't switch	Assess and treat for anaphylaxis
Minor reactions (e.g. minor rash, GI symptoms, myalgias & arthralgias)	YES (be careful that it isn't vasculitis)	YES (be careful that it isn't vasculitis)	Can try switching (but probably not helpful)	Rx anti-histamine for minor rashes

Other Graves' Treatment Options

- Radioactive Iodine (RAI / ^{131}I ...vs RAIU/ diagnostic uptake & scan which uses ^{123}I)
 - Single dose of ablative radioactive iodine.
 - Contraindications: Pregnancy, breastfeeding, **moderate-severe orbitopathy**, thyroid cancer
 - Adverse effects: Worsened Orbitopathy, Thyroiditis
 - Delay pregnancy for 6 months after tx
 - If giving RAI with orbitopathy, should give steroids
 - **Should be off methimazole for *at least* 2-3 days before radioactive iodine ablation**
- Surgery
 - Patient should be euthyroid prior to surgery

Graves' Orbitopathy



Mild

- Artificial tears or ophthalmic gels
- Selenium supplements x 6 months

Moderate to Severe

- IV glucocorticoids + mycophenolate is the EUGOGO first-line treatment for those with **ACTIVE** GO if no contraindications (i.e. CHF, severe hyperglycemia).
- Alternative first-line regimen is IV glucocorticoids alone at higher doses
- Surgery only offered for stable **INACTIVE** GO (must be inactive >6 months)
- Teprotumumab – not yet approved in Canada

Graves' in Pregnancy

1. Anti-thyroidal medications are teratogenic
 - a) PTU in 1st trimester (conversion ~1mg MMZ:20mg PTU), MMZ after that (or discontinue all ATDs if possible)
 - b) Use the lowest possible dose (**aim for T4 at high-normal range**)
2. Check TSH-R Ab in 2nd trimester as can cross placenta
 - if >3x ULN, predicts risk of neonatal hyperthyroidism
 - *Consult Pediatrics to see Newborn*
3. Treatment with **β -blockers** in pregnancy can be associated with lower birth weight/IUGR (more so w long-acting and lipophilic β -blockers such as atenolol, propranolol may not significantly affect BW¹), fetal bradycardia, and neonatal hypoglycemia
4. Watch for post partum exacerbations and post partum thyroiditis

¹ Duan L et al. J Clin Hypertens 2018; 20(11):1557-1649.
ATA Guidelines 2017

Gestational Transient Thyrotoxicosis

- In normal pregnancy, **Thyroid Binding Globulin** and **total T4 increase** by 7 wks GA and **peak at 16 wks GA**
- **hCG** stimulates the TSH receptor on the thyroid gland causing **↑thyroid hormone & ↓TSH**
- The hCG effect may be even more pronounced in: hyperemesis gravidarum, molar pregnancy, multiple gestation, choriocarcinoma
- Generally self-limited, improves by 14-18 weeks
- Use B-blockers if really necessary for symptoms

Thyroid Storm

Think of this with a very sick patient with thyrotoxicosis (tachycardia, confusion, hyperthermia)

The degree of T4 excess is not necessarily more than with other forms of hyperthyroidism

Burch-Wartofsky Scale <https://www.mdcalc.com/burch-wartofsky-point-scale-bwps-thyrotoxicosis>

don't memorize the calculations, know the clinical picture Scale helps identify impending thyroid storm (admit, treat) / severe thyroid storm (ICU) from "unlikely" (workup as outpt)

COMPONENTS:

- Fever
- Neurological symptoms
- GI symptoms/hepatic dysfunction
- Tachycardia
- A. fib +/- Heart failure
- Precipitants (infection, surgery, trauma, iodine load, pregnancy; **most common is medication non-adherence/discontinuation**)

Thyroid Storm Treatment

- ABCs – get ICU involved early!
- Supportive care
- **Beta-blockers** (careful with hemodynamic status!!)
 - E.g. Propranolol 60-80mg PO q4-6h, if unsure, start at lower Propranolol dose 20-40mg po q4-6hr [the intention is to reduce adrenergic drive]
- **PTU*** (usually 200 mg PO q4h) **THEN**
- **Iodine**
 - Lugol's iodine 10 drops q8h
 - Should be given 1 hour **after** the loading dose of PTU
- **Glucocorticoids** (often AI co-exists, also helps to reduce fT4 → fT3 conversion)

Avoid ASA – displaces T4 and T3 from thyroid binding globulin

*In Real Life: PTU is preferred in Thyroid storm vs MMI because it blocks some peripheral conversion of T4 → T3. If unavailable at your hospital (intermittent shortages since 2020 in Canada): give MMI 20mg q4-6h

MCQ 2 - 2026

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- b) Methimazole
- c) **Propylthiouracil (PTU)**
- d) Radioactive iodine (I-131)

C – PTU is preferred to treat Graves' in the first trimester due to lower risk of teratogenicity compared to MTZ. In second trimester patients can be switched to MTZ as it does have lower risk of liver toxicity

4. Hypothyroidism

Resources: 2014 ATA Hypothyroidism Guidelines

<https://www.liebertpub.com/doi/full/10.1089/thy.2014.0028>

Canadian Task Force Guidelines 2019

“ recommends **against screening*** **asymptomatic nonpregnant** adults aged 18 years and older for thyroid dysfunction in primary care settings”

*Screening = measure TSH

<https://canadiantaskforce.ca/guidelines/published-guidelines/asymptomatic-thyroid-dysfunction/>

Subclinical Hypothyroidism

- **TSH** above upper limit of normal with **normal Free T4**

Treat when:

- TSH > 10 mIU/L
 - * 1.6mcg x weight (kg) = typical starting dose; titrate every 4-6 weeks
 - ** **if cardiac disease (CAD)** or frail elderly, can start **at lowest possible dose** (e.g. 25mcg po daily or 50mcg po daily to avoid risk of inducing AFib)

Consider treatment when:

- Symptomatic
- Goiter
- Pregnancy/pregnancy-planning
- Positive anti-TPO antibodies

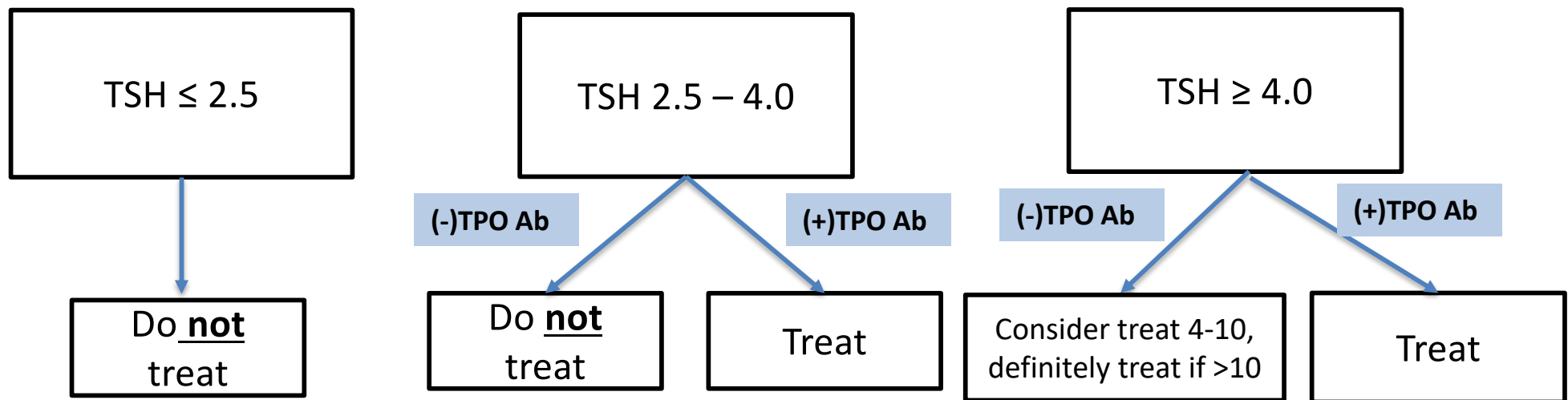
Double-blind RCT of adults > 65 years with subclinical hypothyroidism (Stott DJ et al. NEJM 2017):

No apparent benefits to treating older persons with sub-clinical hypothyroidism

Choosing Wisely: routinely, we do not test for anti-TPO (also called anti-microsomal Ab), but in subclinical, can test to push your treatment decision

Pregnancy: Subclinical Hypothyroidism

Simplified version of the 2017 ATA Guidelines



- ❖ Pregnant women with TSH >2.5 mU/L should be evaluated for TPO-Ab
- ❖ **Once on treatment, target TSH ≤ 2.5 throughout the pregnancy**
- ❖ **In Canada, we do not routinely screen TSH in asymptomatic pregnant women**
- ❖ **If pre-existing hypothyroidism, need ~30-35% more LT4 as soon as pregnant**
i.e. extra LT4 pill on Sat and Sun (9 pills/week)
Upon delivery, go back to pre-pregnancy dose

Myxedema Coma

- Severe hypothyroidism leading to:
 - Altered LOC / lethargy
 - Hypothermia
 - Hypotension
 - Bradycardia
 - Hyponatremia
 - Hypoventilation

Life-threatening SLOWING of function in multiple organs!

Diagnosis is clinical – but often see very low/undetectable FT4, high TSH

Myxedema Coma Treatment

- **IV levothyroxine** load 200-400mcg x1 followed by 1.6mcg/kg/d (this is the PO dose, multiply by 75% if given IV)
 - Lower dose should be considered if cardiac history or elderly
 - PO levothyroxine ~75% as potent as IV form
- **IV glucocorticoids** (HC 100mg IV Q8H) until AI ruled out
- **IV liothyronine** load 5-20mcg x1 followed by 2.5-10mcg Q8H
- **Supportive measures** (ICU monitoring, mechanical ventilation, fluids, warming)

6. Osteoporosis

Highest yield resource:

***NEW 2023 Osteoporosis Canada Guidelines (Scope of Recommendations: Community-dwelling postmenopausal females & males aged 50 years +)

<https://www.cmaj.ca/content/195/39/E1333>

Recommendations for screening for 1o prevention of Fragility Fracture – Canadian Task Force on Preventative Health Care – Theriault G et al. CMAJ 2023 195(18) E 639-E649.

American College of Endocrinology (ACE), the American Association of Clinical Endocrinologists (AACE) guidelines for the diagnosis and treatment of Postmenopausal Osteoporosis (2020)

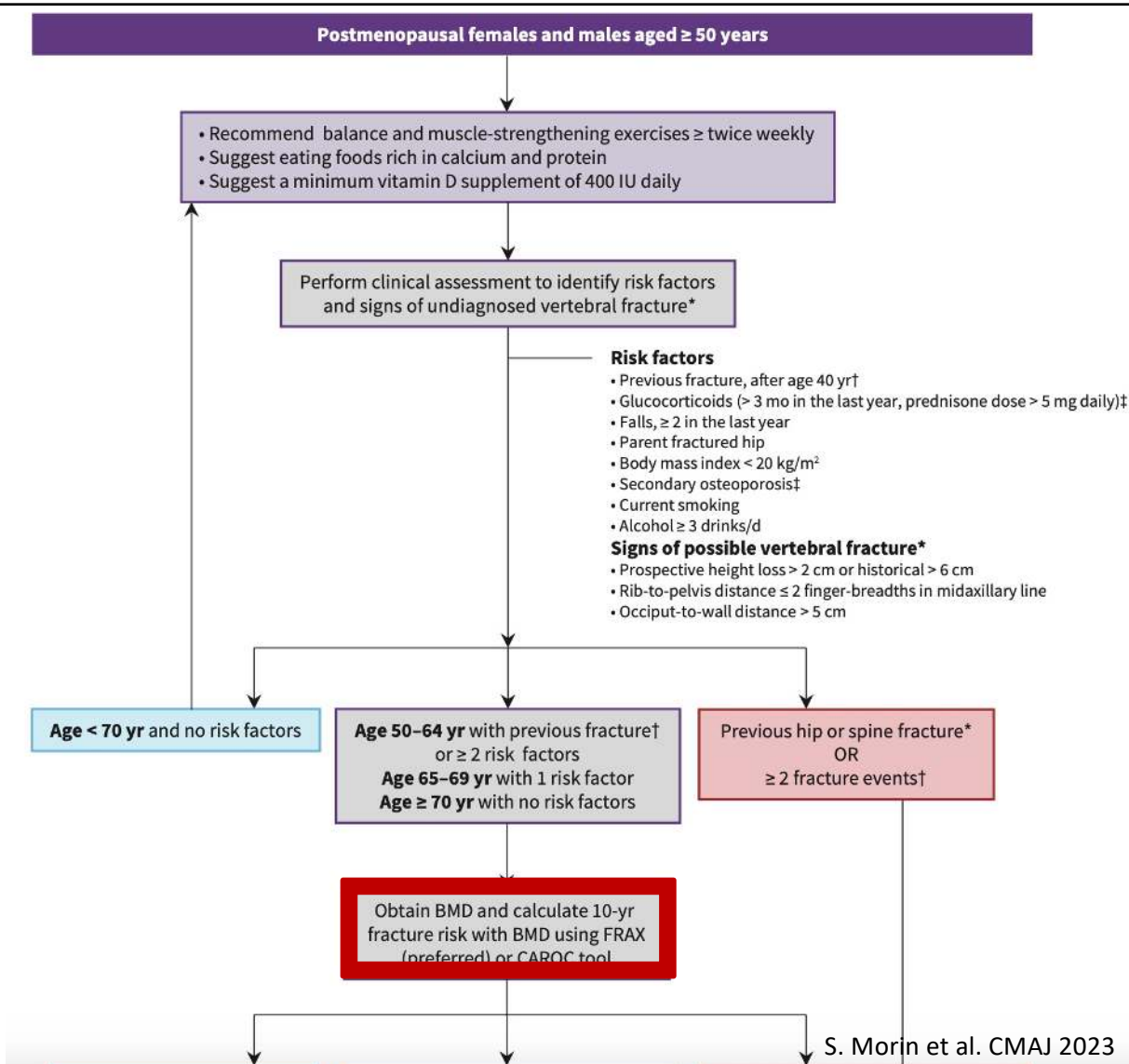
[https://pro.aace.com/sites/default/files/2020-05/Vol%2026%20Supplement%201%20\(May%202020\)%20GL-2019-0524_0.pdf](https://pro.aace.com/sites/default/files/2020-05/Vol%2026%20Supplement%201%20(May%202020)%20GL-2019-0524_0.pdf)

Pharmacological Management of Osteoporosis in Postmenopausal Women(2019)

<https://www.endocrine.org/clinical-practice-guidelines/osteoporosis-in-postmenopausal-women>

Osteoporosis Canada 2023 Guideline

IMR recommends
following these screening
recommendations over
Canadian Task Force



FRAX Tool (<https://frax.shef.ac.uk/>)

Age, BMI

Previous fracture?

Parent with fractured hip?

Current smoking

Glucocorticoids

RA

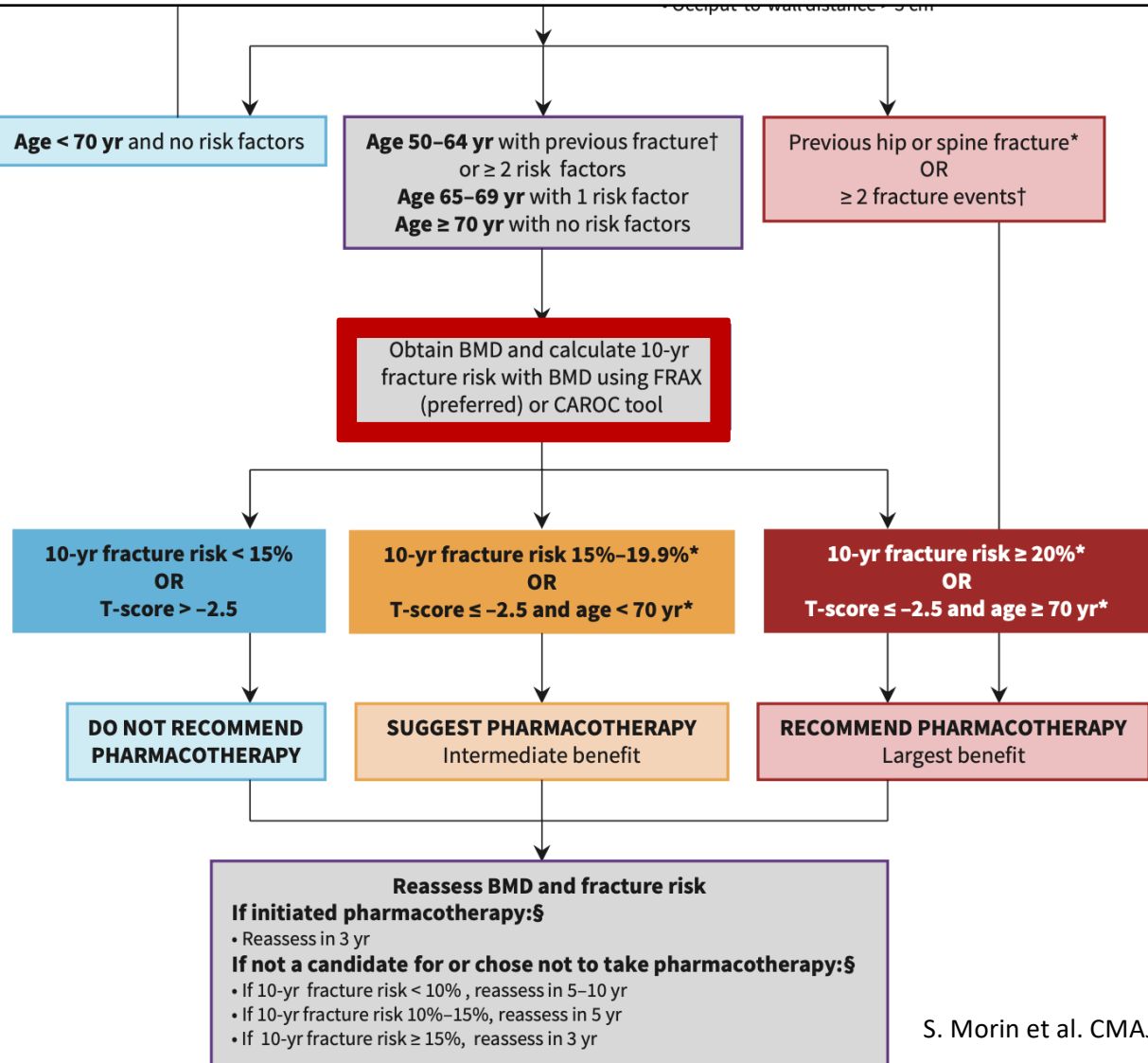
2° osteoporosis - Conditions associated with OP

Type I (insulin dependent) diabetes, osteogenesis imperfecta in adults, untreated long-standing hyperthyroidism, hypogonadism or premature menopause (<45 years), chronic malnutrition, or malabsorption and chronic liver disease

Alcohol 3 or more per day?

Generates a 10-year probability of fragility fracture (%)

FRAX May underestimate risk when very low BMD at spine or total hip, recent fractures, multiple fractures, recurrent falls, other co-morbidities



S. Morin et al. CMAJ 2023

Diagnosis of Osteoporosis

“Bone mineral density of 2.5 or more standard deviations (SD) below the peak bone mass for young adults (*T-score ≤ -2.5) on dual-energy x-ray absorptiometry (DEXA)”

OR Clinically...

*Premenopausal women & men <50y
=should use Z-score instead of T-score

1) **Fragility Fracture**

- Low trauma (e.g. from standing height at walking speed or less, bend, sneeze, cough)
- Major osteoporotic fracture: Fractures of the hip, vertebra, humerus or pelvis (NOT of the hands, feet, craniofacial bones)
- Vertebral fractures, even asymptomatic, associated with $\uparrow$ risk of fractures at all skeletal sites

OR

2) Absolute **fracture risk of 20%** or more over the next 10 years on FRAX or CAROC

FRAX tool is preferred (High risk = >20% major osteoporotic fracture, > 3% hip fracture) superior for classification and treatment <https://www.sheffield.ac.uk/FRAX/tool.aspx?country=19>

CAROC (High risk = >20% fracture) <https://osteoporosis.ca/wp-content/uploads/CAROC.pdf>

Note that either tool could potentially underestimate risk : recency of fractures, recurrent falls, other comorbidities or very low BMD at the lumbar spine and total hip sites

Who to Screen

BMD

- **50–64 yr** with a previous osteoporosis-related fracture or ≥ 2 clinical risk factors
- ≥ 65 yr with **1** clinical risk factor for fracture
- ≥ 70 yr

Spine x-ray / vertebral fracture assessment

- ≥ 65 yr with **T-score ≤ -2.5** (femoral neck, total hip or lumbar spine)
- 10-yr major osteoporotic fracture risk between **15% and 19.9% (moderate)**
- Clinical signs: **vertebral tenderness, height loss** (historical 6 cm, prospective 2cm), **occiput to wall > 5 cm, rib-pelvis < 2 fingers**

Clinical Risk Factors :

- Previous fracture after 40 yr
- Glucocorticoids (> 3 mo in the last year, prednisone dose > 5 mg daily)
- Falls, ≥ 2 in the last year
- Parent fractured hip
- Body mass index < 20 kg/m²
- Secondary osteoporosis.
- Current smoking
- Alcohol ≥ 3 drinks/d

When to Treat: High Risk

1. **Hip, vertebra or ≥ 2 osteoporosis-related fractures** (don't need FRAX)
2. **10-yr major osteoporotic fracture risk $\geq 20\%$**
3. **≥ 70 yo and T-score ≤ -2.5** (femoral neck, total hip or lumbar spine)

○ Before initiating pharmacotherapy, assess for **secondary causes of osteoporosis**:

Routine labs in workup of secondary osteoporosis:

ALP (Paget's), **PTH**, **Calcium/Alb/PO4** (+/- **PTH** if extended lytes abnormal), **Cr/eGFR** (renal disease), **TSH** (hyperthyroidism), **25OH vitamin D** (order if risk factors for deficiency, malabsorption or starting anti-resorptive), **SPEP** (order if vertebral #)

Other considerations:

Meds/Alcohol, Eating disorders, Bariatric Surgery History, testosterone (if suspect hypogonadism), tTG-IGA (Celiac), PTH if hypercalcemia, etc.

Suggest Treatment: Moderate Risk

- 10-yr major osteoporotic fracture risk between 15-19.9%
- < 70 yr and have a T-score ≤ -2.5 (femoral neck, total hip or lumbar spine)

Other treatment decision factors to consider:

- a fracture in the last 2 years (risk of subsequent fracture highest)
- ongoing risk factors
- 2/3 vertebral fractures are painless (silent)
 - Order thoracic/lumbar x-rays for if age >65 with T-score < -2.5 or 10 yr FRAX risk is 15-19.9% or clinical signs – vertebral tenderness, height loss, occiput to wall >5cm, rib/pelvis < 2 finger breadths)

Pharmacotherapy for Osteoporosis

CLASS	Drugs	Side effects	Contraindications	Rx Notes
Bisphosphonate	Alendronate 70mg po q wk Risedronate 5mg po OD or 35 mg po qwk Zoledronic acid (5mg IV annually)	Esophageal / GI intolerance MSK arthralgias Flu-like infusion reaction (IV Zole) Rare: AFF, ONJ	CrCl<30-35 (Oral) CrCl<35 (IV zoledronic acid) Esophageal abnormalities (Oral) HypoCalcemia	Empty stomach (unless DR formulation), stay upright 30-60 min after
Rank-L inhibitor	Denosumab (sc Q6 months)	HypoCa, Dermatitis MSK arthralgias Rare: AFF, ONJ	HypoCalcemia	Adequate Vit D to avoid hypoCa / Caution if severe renal impairment. FDA approved for any GFR Rapid bone loss if D/C**
Hormonal Therapy	HRT	VTE, Cardiac, stroke, breast CA	VTE, active CVD, stroke, E2 dependent tumours, vaginal bleeding, liver dz	Only postmenopausal women
	Raloxifene (SERM)	VTE, cardiac, stroke Vasomotor sx's, Leg cramp	VTE, CVD, stroke, vaginal bleeding	Only postmenopausal women
Parathyroid hormone analog	Teriparatide (20-40ug sc daily)	Orthostatic hypoBP Nausea, HyperCa, hyperCalciuria MSK discomfort	CrCl<30 , Bone Ca Paget's, HyperCa, skeletal rads Unexplained ↑ALP	Caution if nephrolithiasis \$\$\$\$\$
Sclerostin inhibitor	Romosozumab (210mg sc month)	MI, stroke, HypoCa MSK discomfort Rare: AFF, ONJ	Previous Stroke or MI HypoCa	Adequate Vit D to avoid hypoCa, \$\$\$\$\$ Caution if severe CKD

Pharmacotherapy Choices

1. Oral Bisphosphonates (1st line due to cost and access)
2. IV Zoledronic Acid (If GI effects on oral bisphosphonates)
3. Denosumab (If intolerant to bisphosphonates) *

Osteoporosis Canada cautions against using denosumab first line due to **risk of rapid bone loss and vertebral fractures with delayed dosing or discontinuation. Reserve for those intolerant to bisphosphonates or if barriers to access IV bisphosphonates*

4. Romosozumab (12mo) or Teriparatide (24mo)
 - If recent severe vertebral fracture (=height loss >40%) OR > 1 VB fracture AND a T-score ≤ -2.5)
 - one course then **transition to antiresorptive** agent to maintain bone density gains
5. HRT (if female < 60, within 10 yr of menopause start, with goals to treat menopause symptoms Ω)
6. Raloxifene (Over no treatment in postmenopausal females who cannot tolerate any other therapies)

Bisphosphonates: initial therapy for 3–6 yr, Reassess during drug holiday with BMD/FRAX in 3 years (or sooner if clinical indication)

Denosumab:

PLAN long-term uninterrupted therapy (should not delay Q6month dosing by more than 1 mo) may be re-assessed after 6–10 yr

If stop at ≤ 4 doses, **transition to bisphosphonate 6 mo after the last dose of denosumab x 1 yr** then re-assess

If stop at ≥ 5 doses (**risk of rapid bone loss or vertebral fractures is high**): refer to OP expert

BONUS
Read on own

BMD Monitoring Frequency

- Not a candidate for pharmacological treatment:
 - 5–10 yr: if risk of major osteoporotic fracture (MOF) is < 10%
 - 5 yr if risk of MOF is 10%–15%
 - 3 yr if the risk of MOF is > 15%
- Those on Therapy:
 - 3 yr after starting Rx
- Those on Bisphosphonate Drug Holiday:
 - 3 yrs after stopping

*BMD measurement may be repeated **sooner/more frequently** in people with new fracture, secondary causes of osteoporosis, or new clinical risk factors associated with rapid bone loss

Atypical Femoral Fractures (AFF)

Thigh or groin pain on bisphosphonates = need to rule out AFF! (bilateral femur X-rays, if these negative consider bone scan / MRI if high index suspicion)**



- Radiographic Characteristics
 - Lateral cortical thickening
 - Transverse fracture lines
 - Beaking

- Proximal femoral shaft fracture
- Atraumatic
- Chronic bisphosphonate use (risk increases as early as 3 years)
- Asian women high risk
- Prodromal thigh pain (in up to 70%)
- **Can occur with denosumab and romosozumab as well**

Absolute risk increase is small, ~1:1000 patient years on bisphosphonate, versus 1:50000 patient years off bisphosphonate

Management of AFF

PREVENTION

- Drug holiday for oral/?IV bisphosphonate for 3-6 years
- AFF risk is lower 2 years after stopping drug

TREATMENT

- Orthopedic surgery consult (for IM nail insertion)
- **Image contralateral femur** (early fractures may be asymptomatic)
- Stop bisphosphonate
- Ensure adequate vitamin D and calcium intake, address other clinical risk factors for falls and fractures
- **Start teriparatide (if still meet OP treatment criteria)**
 - **Denosumab NOT a good option as still risk of AFF** - risk of AFF stable over time whereas with bisphosphonates the risk increases with time

Osteonecrosis of the Jaw ONJ

- Nonhealing wound in oral mucosa with exposed bone that lasts >8 weeks
- Risk <1/100,000 patient-years
- Dental care
- Increased risk if jaw surgery



My patient had a hip fracture on treatment... now what?

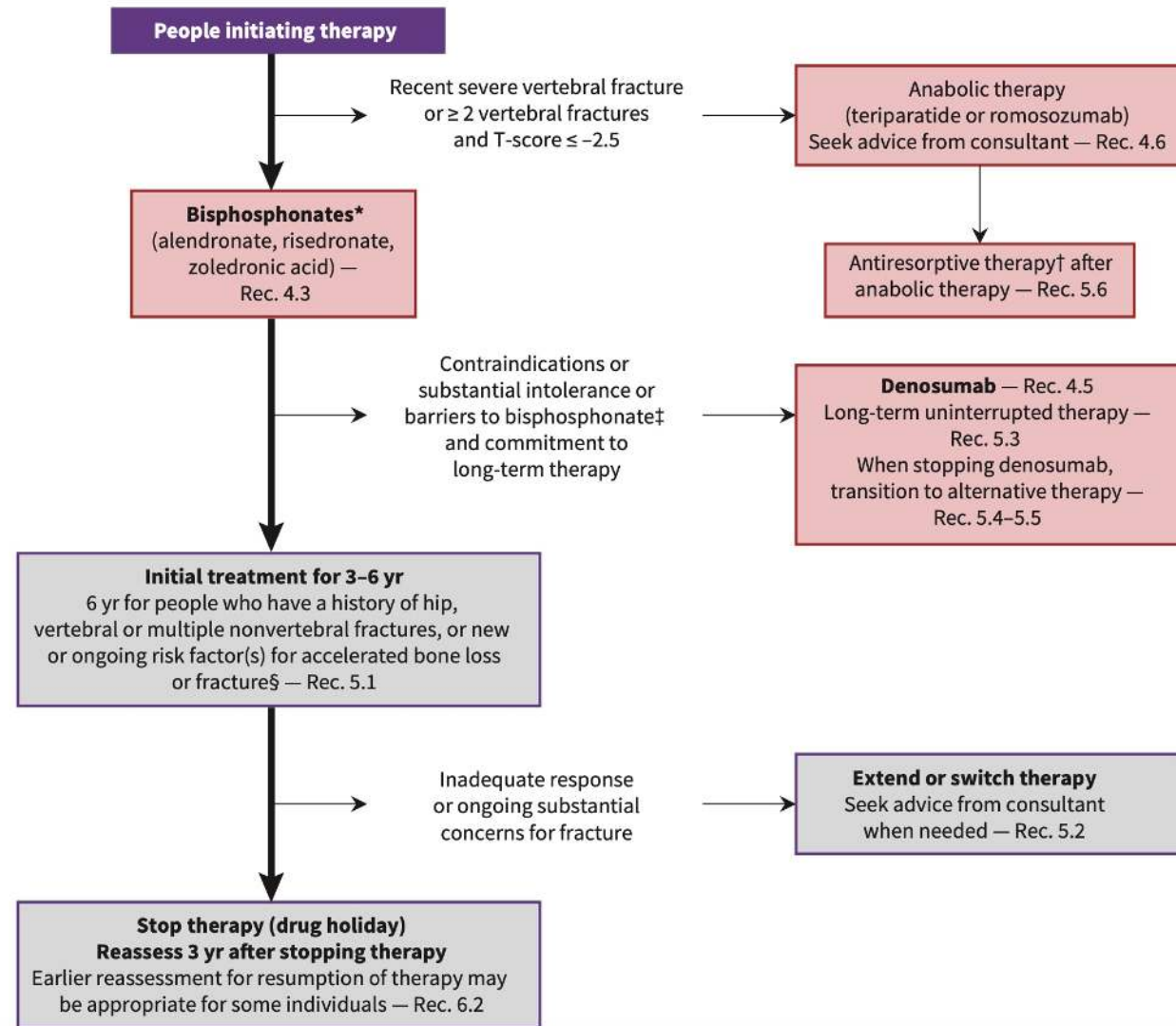
Inadequate response

- **> 1 fracture** or substantial **bone density decline** (e.g. $\geq 5\%$)
 - Despite adherence to an adequate course of treatment (typically >1 yr)
- Consider: 2^o causes of osteoporosis, falls, BMD imprecision errors
- Do not use bone turnover markers or FRAX/CAROC for monitoring

What to do:

- extend or switch therapy
- reassess for secondary causes and refer to OP expert

Treatment Summary



Evidence based Non-Pharm Interventions

Exercise



- Functional & Balance Training (twice weekly) : ↓ fall related fractures
- Resistance Training (twice weekly): In combination with functional
- **Caution:** Twisting or Flexion of the spine in repetitive, rapid way

Nutrition



- Vitamin D: RDI age <70 = 600 IU, age >70 = 800 IU
 - Supplement Vit D3 - dosing to target 25-OH vitamin D level *75- 125 nmol/L
- Calcium: 1000 mg/d (males aged 51–70 yr) and 1200 mg/d (females > 50 yr & males > 70 yr), dietary sources preferred
- Supplemental Protein, vitamin K, magnesium: low evidence

Other

- Smoking cessation, alcohol moderation

BONUS
Read on own

Paget's Disease of Bone

- **Pathophysiology:** focal $\uparrow$ in bone resorption by very large osteoclasts, then $\uparrow$ osteoblastic activity producing a high rate of bone formation/turn-over.
- Initial Labs: $\uparrow$ serum total alkaline phosphatase (ALP) +/- bone specific ALP, without other abnormalities
- Imaging: **do plain film XRs first of suspicious areas, if asymptomatic do skeletal survey/series (look for thickened cortices with tunnelling & accentuated trabeculae)**
- **If the diagnosis is confirmed $\rightarrow$ bone scan** should be done to determine the extent of the disease.
- Associated illness:
 - **Hearing loss, compressive neuropathies, osteoarthritis, osteosarcoma**
- Indications for Treatment:
 - Symptoms – pain, compression, fracture
 - Evidence of active disease with impending symptoms (high risk for fracture)
 - Hypercalcemia (usually should only happen if patient is immobile)
 - ALP $>2\times$ ULN
 - Pre-orthopedic surgery at or near site (i.e. femoral Paget's lesion for hip replacement)
- First Line Treatment: IV zoledronic acid 5mg IV q1yr is treatment of choice
or Oral bisphosphonate can be used ****NOTE DOSING****
 - alendronate 40 mg/day x 6 months
 - Risedronate 30mg/day x 2 months
- **2nd Line Agent (if intolerant to bisphos) is Calcitonin**

Paget's Disease of Bone:
An Endocrine Society Clinical
Practice Guideline (2014)

7. Diabetes

Highest yield resource: **2018 Diabetes Canada Guidelines**

<http://guidelines.diabetes.ca/cpg>

+ **2020 Diabetes Canada Updates:**

<http://guidelines.diabetes.ca/2020-Update> (Pharmacologic glycemic management of T2DM in adults)

+ **2021 Diabetes Canada Updates:**

<http://guidelines.diabetes.ca/2021-Update> (BG monitoring in adults & children with diabetes)

+ **2022 Diabetes Canada Updates:**

<https://guidelines.diabetes.ca/2022-Update> (T2DM Remission)

+ **2023 Diabetes Canada Updates:**

<https://guidelines.diabetes.ca/2023-Update> (Diabetes and mental health)

OTHER – KDIGO CKD and Diabetes Guidelines 2023

Note: The chapter summaries are concise.

Diabetes - Diagnosis

Diagnosis of Diabetes with one of:

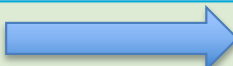
- Fasting glucose $\geq 7\text{mmol/L}$ (Prediabetes: 6.1-6.9) OR
- HbA1c $\geq 6.5\%$ (Prediabetes: 6.0-6.4%) OR
- 2h 75g OGTT $\geq 11.1\text{ mmol/L}$ (Impaired glucose tolerance: 7.8-11.0) OR
- Random glucose $\geq 11.1\text{ mmol/L}$

Type of Diabetes:

- **Type1:** insulin deficiency (pancreatic β cell destruction), DKA prone, Usually Anti-GAD/Anti-Islet Cell antibody +
- **Type 2:** insulin resistance (tissue resistance, secretory defect), metabolic syndrome
- **Latent autoimmune diabetes in adults (LADA):** Type 1.5 diabetes with immune-mediated loss of pancreatic β cells
- **Monogenic:** familial autosomal dominant mutation leading to beta cell defects, neonatal or <25 yo
- **GDM:** glucose intolerance during pregnancy
- **Pancreatic Insufficiency** – (eg) CF, Chronic Pancreatitis, 2o to Hemochromatosis, post Whipple etc.

A1C Treatment Targets

A1C% Targets

<6.0	Selected adults with type 2 diabetes with potential for remission to normoglycemia
≤6.5*	Adults with type 2 diabetes to reduce the risk of chronic kidney disease and retinopathy if at low risk of hypoglycemia [†]
≤7.0	MOST ADULTS WITH TYPE 1 OR TYPE 2 DIABETES 
7.1  8.5	7.1-8.0%: Functionally dependent [†] 7.1-8.5%: • Recurrent severe hypoglycemia and/or hypoglycemia unawareness • Limited life expectancy • Frail elderly and/or with dementia[‡]

<7% corresponds to:
 FBG 4.0-7.0
 2h pc BG 5.0-10.0

Avoid higher A1C to minimize risk of symptomatic hyperglycemia and acute and chronic complications

End of life: A1C measurement not recommended. Avoid symptomatic hyperglycemia and any hypoglycemia.

* Target 6.0 to <6.5 for adults with type 2 diabetes with potential for remission to prediabetes

[†] Based on class of antihyperglycemic medication(s) utilized and the person's characteristics

[‡] See Diabetes in Older People chapter

BONUS

Read on own

A1C confounders

Factors affecting A1C	Increased A1C	Decreased A1C	Variable Change in A1C
Erythropoiesis	B12/Fe deficiency Decreased erythropoiesis	Use of EPO, Fe, or B12 Reticulocytosis Chronic liver disease	
Altered hemoglobin			Fetal hemoglobin Hemoglobinopathies Methemoglobin
Altered glycation	Chronic renal failure ↓erythrocyte pH	ASA, vitamin C/E Hemoglobinopathies ↑ erythrocyte pH	
Erythrocyte destruction	Splenectomy	Hemoglobinopathies Chronic renal failure Splenomegaly Rheumatoid arthritis HAART meds, Ribavirin Dapsone	
Assays	Hyperbilirubinemia Carbamylated Hb ETOH Chronic opiates	Hypertriglyceridemia	

Type 1 Diabetes

Management



T1DM Management

- Basal-Bolus Injection (BBI)
- Continuous Subcutaneous Insulin Infusion (i.e. insulin pump therapy)
- +/- Continuous Glucose Monitoring (CGM)
 - CGM Benefits: improves A1C, reduces hypoglycemia, provides detailed glucose trend data for better management, yields greater benefits with consistent use (>70% adherence)
 - No recommendations for non-insulin antihyperglycemic agents in T1DM
 - Guideline reports no long-term benefit of metformin added to insulin in obese T1DM
 - Caution re DKA warranting more research for SGLT2 inhibitors, GLP-1 receptor agonists
 - not recommended for use at this time
- Screen for auto-immune conditions: Check TSH, anti-TPO Ab at diagnosis and q2-5 years thereafter, adrenal insufficiency if symptoms, celiac disease if symptoms i.e. iron deficiency, unexplained hypoglycemia, vitamin D deficiency, osteoporosis

BONUS
Read on own

Continuous glucose monitor (CGM) – Targets

Glucose parameter	Normal adults	Older / High Risk for Hypoglycemia	Pregnant T1DM
Time in Range (Target Range)	70% (3.9-10 mmol/L)	50% (3.9-10 mmol/L)	70% (3.5-7.8 mmol/L)
Time Below Range	<4% (<1% under 3.0)	<1%	<4% (<1% under 3.0)
Time Above Range (TAR)	<25%	10.1-13.9 = acceptable at times >=14 = <10% TAR	<25%
NOTES	70% TIR achieves an A1C close to 7%	50% TIR equates to an A1C of about 8.0%. Each 10% TIR equates to about 0.5-0.8% change in A1C	Targets for T2DM in pregnancy and GDM not yet known (insufficient data)

MDI (Multiple Daily Injections) Regimen Tips

- High glucose? Look at the dose before...
 - High fasting glucose? ↑ bedtime basal
 - High pre-lunch glucose? ↑ breakfast bolus
 - Exception is GDM where targets are post-prandial, so look at the bolus at that meal
- Nocturnal Hypoglycemia
 - May lower basal
 - Use ultra long acting basal (e.g. Tresiba (insulin degludec) – Note make dose changes every 2-3 days
 - Long-acting insulin recommended over NPH in T1DM to avoid hypoglycemia
 - May introduce a mixed meal bedtime snack
 - Obtain history re: dinner and dinner bolus

- Ratio Basal-bolus
 - Individualized, there's no recommended ratio
 - e.g. 50:50, 60:40, 40: 60

Some Math

Insulin Sensitivity: $100/\text{total daily insulin dose}$

Carb ratio : $500/\text{total daily insulin dose}$

- In Hospital Situations
 - Targets BG <12
 - Glucocorticoids, TPN

Once-weekly insulins (e.g., insulin icodec “Awiqli”) are ultra–long-acting basal insulins reduce injection burden and maintain stable glucose, \$\$\$ costly

DKA : Diagnosis and Treatment

DKA Diagnosis: AGMA

Arterial pH is ≤ 7.3

- bicarbonate ≤ 15 mmol/L
- anion gap is >12 mmol/L
- +Ketones (serum/urine)

Glucose is usually ≥ 14.0 mmol/L

- can see 'euglycemic DKA' in profound fasting or concomitant use of SGLT2i

vs. HHS: minimal to no acidosis

- Usually Glucose ≥ 34.0 mmol/L
- Increased plasma osmolality >320 mOsm/kg
- greater ECFV contraction
- Neuro: more decreased LOC, seizures

+ WORKUP for CAUSE – The “5 I”s

Intoxication, Infarction (MI/Stroke), Insulin deficiency (missed doses), Infection, Iatrogenic

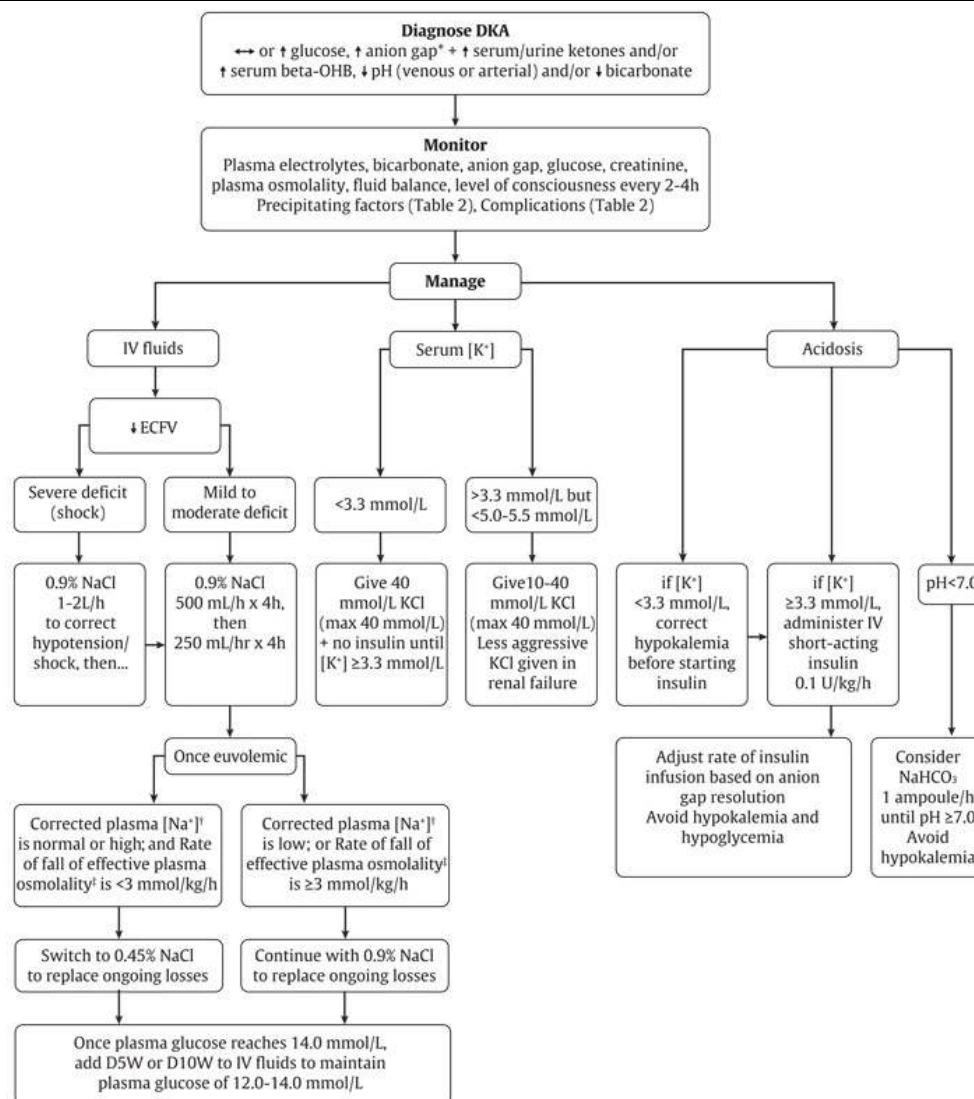
When Anion Gap corrects – and patient feeling better –
Transition to sc insulin: 2-3 h overlap IV gtt after giving sc insulin, ensure patient is eating

DKA Management: 3 Pillars

- 1) **Watch Potassium carefully, replace hypokalemia**
 - If K <3.3 , replace potassium before giving IV insulin or IV bicarb! *Both will shift K intracellular, worsening hypokalemia!*
 - Usually K is high at presentation (due to shifting extracellular H $^{+}$ $\leftrightarrow$ for intracellular K $^{+}$)
 - Once K falls to <5.5 can add IV KCl to maintenance fluid
- 2) **Treat volume contraction** with isotonic crystalloid (0.9% NS). Switch to hypotonic fluid (e.g. 0.45% NaCl with 40meq/L KCl once not in hypovolemic shock)
 - Excessive NS can cause a hyperchloremic metabolic acidosis (NAGMA)
- 3) **Treat ACIDOSIS with IV Insulin**
 - Bolus 0.1U /kg then run infusion 0.1U/kg/hr per standardized protocols
 - When Glucose is <13 add DEXTROSE to maintenance fluid – you are correcting ACIDOSIS (anion gap) not just the blood glucose
 - You will have to do this EARLY in euglycemic DKA on SGLT2i
 - Very limited role for IV bicarbonate

BONUS
Read on own

DKA



vs. HHS:

- Usually BG ≥ 34.0 mmol/L
- Increased plasma osmolality > 320 mOsm/kg
- greater ECFV contraction, minimal to no acidosis
- Neuro: more decreased LOC, seizures

Type 2 Diabetes

Management



“Resistance training is just as important as cardio. Train yourself to resist chocolate, pastries, fried foods, beer, pizza....”

MCQ 3 - 2026

A 72-year-old man with type 2 diabetes for 12 years is on metformin 1000 mg BID. His HbA1c is 8.9%. He has a BMI of 38 kg/m² and reports difficulty with weight management. He has mild non-proliferative diabetic retinopathy on exam but no history of ASCVD, CKD, or heart failure. eGFR is 88 mL/min. He is motivated for weight loss and wants to minimize hypoglycemia risk.

Which of the following is the most appropriate next agent to add?

- a) Basal insulin
- b) Semaglutide
- c) Tirzepatide
- d) Gliclazide
- e) Empagliflozin

T2DM Treatment: At Time of Diagnosis

- Non-pharmacological management
 - refer to Diabetes Education Centre, set A1C target, assess CV status, smoking cessation etc
 - Diet/Lifestyle changes for all
- **Metformin if A1C $\geq$ 1.5% above target at diagnosis**
 - Insulin + Metformin if *symptomatic hyperglycemia and/or metabolic decompensation* 
- Reassess in 3-6 months

Symptomatic hyperglycemia with metabolic decompensation:

Suggests a state of relative or absolute insulin deficiency.

- Polyuria
- Polydipsia
- Blurry Vision
- Unintentional Weight loss
- Volume depletion
- Ketosis
- HHS/DKA

SHOULD BE STARTED ON INSULIN +/- metformin until glycemic control is achieved! (then you back off insulin if you can, while adding on oral agents)

Summary of Current Evidence for Type 2 DM

A1C >1.5% above target at diagnosis and no metabolic decompensation

Frail or no risk factors

Either of:
- Stroke
- Obesity

HFnrEF

HFrEF

Any of:
- Clinical CVD
- CKD with $\uparrow$ ACR
- *CV risk factor
- HFpEF

Metformin only
- Can add DPP4i, individualize

Metformin + GLP1-RA

Metformin + SGLT2i + GLP1-RA*
with LVEF $\geq$ 45% and BMI $\geq$ 30 (CCS HFnrEF 2025)

Metformin + SGLT2i

Metformin + SGLT2i / GLP1-RA
- if not at target add the other

*CV risk factors:

- Smoker
- HTN (>140/95 or on meds)
- Dyslipidemia (on statin, or baseline LDL >3.4 or HDL <1(M) / 1.3(F) or TG >2.3)
- Central obesity

Adapted from Cheng AY et al. Can J Diabetes 2020; 44:1-3.

Summary Data for GLP1-RA

not yet reflected in 2020/2022 CDA guidelines

✓ CVD history

Semaglutide + obesity

= ↓ CV death, MI, Stroke

Liraglutide + T2DM + obesity

= ↓ CV death, ↓ all-cause death

Dulaglutide + t2DM

= ↓ MACE (non significant for death)

✓ CKD history

Semaglutide + T2DM + obesity

= ↓ CV death, ↓ all-cause death ↓ CKD progression

✓ HFpEF (HFnrEF) history

Semaglutide + T2DM + obesity

= ↓ HF, ↓ A1C, ↑ exercise capacity

Tirzepatide + obesity

= ↓ CV death, ↓ HF

HFrEF history

Liraglutide + admitted w HF

= ↑ **HF events** [FIGHT 2016, LIVE 2017]

No semaglutide trials

Note that exenatide, lixisenatide have no proven cardiorenal benefits, don't pick on MCQ

N Engl J Med. 2024 Apr 18;390(15):1394-1407
 N Engl J Med. 2024 Jul 11;391(2):109-121
 N Engl J Med. 2023 Dec 14;389(24):2221-2232
 JAMA 2016; 316(5): 500-508
 Eur J Heart Fail. 2017; 19(1): 69-77. 68

Summary Data for SGLT2i

- ✓ **CVD** history
 - canagliflozin + t2DM = ↓ MACE (but trend to more lower limb amputations)
 - dapagliflozin + T2DM = ↓ HF hospitalization
 - Empagliflozin + T2DM = ↓ CV death, ↓ HF hospitalization and renal events
- ✓ **CKD** history
 - canagliflozin + t2DM + GFR>30 = ↓ CKD progression, ↓ CV events
 - dapagliflozin +/- T2DM + GFR>25 + ACR>200 = ↓ CKD progression, ↓ all cause mortality
 - Empagliflozin +/- T2DM + GFR>20+ ACR>200 = ↓ CKD progression, ↓ CV death
- ✓ **HFpEF (HFnrEF)** history
 - dapagliflozin +/- T2DM = ↓ HF hospitalization/events
 - Empagliflozin +/- T2DM = ↓ HF hospitalization/events
- ✓ **HFrEF** history
 - dapagliflozin +/- T2DM = ↓ CV death, ↓ HF hospitalization/events
 - Empagliflozin +/- T2DM = ↓ CV death, ↓ HF hospitalization/events

Dealer's Choice

MCQs... If asked "which will **reduce CV mortality** in this diabetic patient..." pick SGLT2i.

Domain	SGLT2i (D=dapa, E= empa)	GLP1RA (most data semaglutide)	GIP/GLP1RA (Tirzepatide)
CKD	↓renal failure (D+E) ↓CV death (D+E) ↓all cause mortality (D)	↓renal failure in T2DM w CKD	Signals only (albuminuria, eGFR slope); no outcome trial yet
HFpEF / HFnrEF	↓HF hospitalizations (D+E)	↓symptoms, ↑QoL, pooled analyses show ↓ HF events	↓ HF events
HFrEF	↓HF hospitalizations (D+E) ↓CV death (D+E)	No robust HFrEF outcome benefit	No large HFrEF RCTs published
CVD (MACE)	↓CV death (E) ↓CV death if CKD or HF (D+E)	↓ MACE, ↓ CV events, mortality signals, ↓ stroke	CV outcome trial ongoing (SURPASS-CVOT)
T2DM	↓A1C (0.5-0.8%)	Robust A1c lowering	Very potent A1c lowering
Obesity ↓wt	Modest (~2–3%)	Significant (~15% at obesity dose)	Largest (~20%+)
Mortality	↓ CV mortality (D+E) ↓ all-cause (D+E with CKD)	Semaglutide: Reduced MACE but not CV death alone. Liraglutide ↓ CV mortality in pts w T2DM and high CV risk.	Not yet established

Antihyperglycemic agents & Kidney Function

eGFR	Metformin	SGLT2i	GLP1RA	DPP4i	All Insulins	Secretagogues Glyburide Others	
45-59	2g	No dose change	No dose change	No dose change	No dose change	Avoid	No dose change
30-44	1g	Cana 100mg Dapa 10mg Empa 10mg or 25mg		Linagliptin 5mg Sitagliptin 50mg Saxa 2.5mg*			Dose reduction may be needed
15-29	500 mg			Lina 5mg Sita 50mg			
<15 or on dialysis	avoid	Stop on dialysis	Limited data available				
Risk related to low GFR	Lactic acidosis	Cardiorenal protection preserved but minimal reduction in a1c if GFR <45		Accumulation**	Accumulation and hypoglycemia	Prolonged and severe hypo-glycemia	hypoglycemia

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SGLT2i: Prescribing Notes

CLASS EFFECTS: SGLT2	DRUG/CLASS	PRESCRIBING NOTES
- Risk of euglycemic DKA - GU infections - Yeast infection - Possible perineal necrotizing fasciitis (Fournier's Gangrene) - CKD patients: Glycemic lowering less at lower GFR (but cardiorenal protection preserved) Counsel all patients on "sick day med list" and avoiding very low carb diet, good foot care.	Empagliflozin (SGLT2i)	Ok to start 10mg OD if GFR >20 (per EMPA-KIDNEY Nov 2022)
	Canagliflozin (SGLT2i)	- Ok to start if GFR >30 ↑ risk of toe amputations (noted in CANVAS but not in CREDENCE) - Signal NOT seen for Empa or Dapa)
	Dapagliflozin (SGLT2i)	- Ok to start 10mg OD if GFR >25 - Avoid if active bladder cancer
	Ertugliflozin (SGLT2i)	- Noninferior but not SUPERIOR to placebo for outcome of MACE; do not pick this if given other options that may IMPROVE MACE / CV outcomes. - Longer half life – hold 4 d prior to major surgery

GLP1RA and GIP/GLP1RA prescribing notes

CLASS EFFECTS: GLP1 - RA	DRUG/CLASS	PRESCRIBING NOTES
- Weight Loss - GI side effects : dyspepsia, anorexia, nausea, vomiting, abdo pain, diarrhea - Caution if hx pancreatitis or pancreatic cancer - Contraindicated with personal/family history of medullary thyroid cancer or MEN2*	Semaglutide (SC WEEKLY injection, or oral)	- Worsening retinopathy:SUSTAIN-6 trial **worsening retinopathy may be from <i>rapid</i> glucose lowering** - Also approved in Canada for weight loss without T2DM
	Liraglutide (SC DAILY injection)	Also approved in Canada as a weight loss WITHOUT T2DM
	Dulaglutide (SC WEEKLY injection)	Approved for diabetes only
	Exenatide (SC WEEKLY injection)	CV safety, but NO proven Cardiorenal benefit (<i>don't pick if given other options in same class</i>)
	Tirzepatide **DUAL GIP and GLP1R agonist (SC WEEKLY INJECTION)	Superior to GLP1RA alone for weight and A1C reductions, and non-inferior for MACE. Publication of SURPASS-CVOT trial vs dulaglutide pending.

*New study: GLP1-RA did not increase thyroid cancer risk compared to DPP4i (FU of ~4 yrs)

Diabetes and Macrovascular and Microvascular Protection



Diabetes and Vascular Protection

ABCDE's of Diabetes Management

- **A1C** $\leq 7.0\%$
- **BP** $< 130/80$
- **Cholesterol LDL** $< 2.0^*$ **or** $> 50\%$ reduction
- **Drugs** to protect the heart when indicated → statin, ACEi/ARB, ASA, antihyperglycemic agents with CV benefit (GLP-1 agonist, SGLT2 inhibitor)
 - **NEW** role for non steroidal MRA (finerenone) in T2DM w/ CKD 2024
- **Exercise/Healthy Eating**
- **Smoking cessation**

*1.8 in those with CAD, Stroke history

Drugs for Vascular Protection – Indications

Cholesterol = Statin therapy

- Clinical CVD
- Age ≥ 40
- Age >30 and diabetes duration > 15 years
- Microvascular disease
- Other CV risk factors

Note: **If LDL is not at target with statin therapy alone**, a second line agent may be used.

In patients with clinical CVD, **ezetimibe or evolocumab may be added** to reduce CV events (IMPROVE-IT and FOURIER trials). **SEE CCS LIPID UPDATES**

Drugs for Vascular Protection – Indications

ACEi/ARB

- Clinical CVD
- Age ≥ 55 with an additional CV risk factor or end organ damage (albuminuria, retinopathy, LVH)
- Microvascular disease

Note: Should use doses shown to provide CV protection (perindopril 8 mg daily, ramipril 10 mg daily, telmisartan 80 mg daily)

ASA

- Established CV disease only
- We **no longer recommend** ASA for primary prevention of CVD in people with diabetes (10).

Microvascular Complications

		When to start screening?	How to screen?	Frequency?	Treatment?
Retinopathy	T1DM	5y after diagnosis > age 15	Dilated fundoscopy, fundus photography by optometrist	If retinopathy present – screen per Ophtho If no retinopathy: q1y (T1DM) q1-2y (T2DM)	Tight BG + BP control. Laser therapy, intraocular therapy (anti-VEGF), surgery (vitrectomy)
	T2DM	At diagnosis			
Neuropathy	T1DM	5y of post-pubertal T1DM	10g monofilament test OR 128 Hz Vibration	Annual	Tight BG cntrl For Pain: Anticonvulsants > Antidepressants > Opioid > topical
	T2DM	At time of diagnosis			
Nephropathy	T1DM	5y after diagnosis of T1DM	Random Urine ACR >20, OR 2 of 3 rpt ACRs >2mg/mmol and/or eGFR <60	Annual	• ACE or ARB first line • SGLT2i • Finerenone*
	T2DM	At time of diagnosis			

Microvascular Complications - Neuropathy

Distal symmetric Polyneuropathy most common (pain with numbness, tingling, loss of sensation and ultimately motor involvement (Charcot foot))

Diabetic Autonomic Neuropathy (DAN)

- Cardiac (Cardiac Autonomic Neuropathy) – postural hypotension, resting tachycardia
- GI tract (gastroparesis) and Bladder dysfunction (overflow incontinence), Erectile Dysfunction

Less common: **Isolated neuropathies (eg carpal tunnel) and Diabetic Amyotrophy**

Annual Screening for peripheral neuropathy (10g monofilament test OR 128 Hz Vibration) :

Start after 5 years' duration of post-pubertal Type 1 diabetes

All patients with Type 2 diabetes

Treatment:

1. Intensify glycemic control
 2. Painful Neuropathy: Anticonvulsants (pregabalin, gabapentin, valproate)
- Antidepressants (amitriptyline, duloxetine, venlafaxine)
- Topical Nitrates

Opioids if pain refractory to all of the above, with caution.

See Neuro Lecture –
for workup for peripheral
neuropathy

Microvascular Complications - Nephropathy

Annual Screening:

5 years after diagnosis of Type 1 Diabetes

At diagnosis of Type 2 Diabetes

Dx: Random Urine ACr >20mg/mmol

OR eGFR <60 OR >2 mg/mmol on ≥ 2 of subsequent 3 tests)

Refer if: chronic progressive loss of renal function, ACR persistently >60, eGFR <30, inability to continue on ARB/ACE, unable to achieve target BP

Treatment to reduce risk of progression of Nephropathy:

ACE inhibitor or ARB maximally tolerated dose **DM1, DM2**



Nonsteroidal MRA (finerenone) used in **DM2** + GFR >25 + ACR >3 despite maximally tolerated RASi [KDIGO CKD 2024 Guidelines]

- ↓ progression CKD, ↓ CV and renal death, ↓ AFib (CKD w T2DM)^{1,2}
- *watch out for K levels. If >5.5 → consider binder
- May use steroidal MRA (aldosterone, eplerenone) if CHF, Conn's, refractory HTN with caution in reduced GFR

Approach to Hypoglycemia

Confirm with Whipple's Triad:

1. Symptoms of hypoglycemia
2. Low plasma glucose when symptoms are present
3. Relief of those symptoms with treatment

Some Treatment Options :

- 15 g glucose (Tabs)
- 150 mL juice/Soft Drink
- 15 mL Honey
- Baqsimi 3 mg intranasal
- Glucagon 1 mg IM
- Driving Counseling/Reporting severe hypos – “5 to drive”!
- Review insulin / oral hypoglycemic doses

Evaluation of hypoglycemia in patients not on hypoglycemic agents:

48 - 72 hour fast with capillary glucose monitoring until glucose drops (<3.3 mmol/L), increase to q1hr bloodwork

If venous glucose confirmed < 3 mmol/L, send samples for diagnostic workup (next slide)

Hypoglycemia Workup – Interpreting results

Diagnosis	Insulin (pmol/L)	Proinsulin (pmol/L)	C-peptide (nmol/L)	B-OH butyrate (mmol/L)	Glucose (mmol/L) incr. after Glucagon response	Insulin Antibody
Normal	<21	<5	<0.2	>2.7	<1.4	Negative
Exogenous insulin	>>21	<5	<0.2	<2.7	>1.4	Neg/Pos
Insulinoma, post gastric bypass hypoBG	>=21	>=5	>=0.2	<2.7	>1.4	Negative
Sulfonylurea (SU) overdose	>>21 SU level ++	>=5	>=0.2	<2.7	>1.4	Negative
Insulin autoimmune Ab	>>21	>>5	>>0.2	<2.7	>1.4	Positive
IGF2	<21	<5	<0.2	<2.7	>1.4	Negative
Not insulin mediated (starvation/ malnutrition)	<21	<5	<0.2	>2.7	<1.4	Negative

Diabetes in Pregnancy

Please see **OB Medicine lecture** which will cover GDM, T1DM and T2DM in pregnancy

<http://guidelines.diabetes.ca/cpg/chapter36>

MCQ 3 - 2026

A 72-year-old man with type 2 diabetes for 12 years is on metformin 1000 mg BID. His HbA1c is 8.9%. He has a BMI of 38 kg/m² and reports difficulty with weight management. He has mild non-proliferative diabetic retinopathy on exam but no history of ASCVD, CKD, or heart failure. eGFR is 88 mL/min. He is motivated for weight loss and wants to minimize hypoglycemia risk.

Which of the following is the most appropriate next agent to add?

- a) Basal insulin
- b) Semaglutide
- c) **Tirzepatide**
- d) Gliclazide
- e) Empagliflozin

Insulin and gliclazide effective for A1c reduction but also associated with weight gain and hypoglycemia

SGLT2 inhibitors have cardiovascular and renal benefits, but this patient has neither ASCVD nor CKD. While still reasonable, **GLP-1 RA is prioritized here** for weight loss and glycemic effect. Newer trials show tirzepatide is more effective for weight loss than semaglutide.

8. Adrenal & Pituitary

Highest yield resources:

- CHEP Endocrine Hypertension (hyperaldosteronism, pheochromocytoma):
<https://guidelines.hypertension.ca/diagnosis-assessment/xviii-assessment-for-endocrine-hypertension/>
- 2016 Endocrine Society Adrenal Insufficiency Guidelines
(<https://academic.oup.com/jcem/article/101/2/364/2810222>)
- Adrenal Incidentalomas (2010 and 2020 update):
<https://academic.oup.com/jcem/article/95/9/4106/2835193>
- <https://academic.oup.com/edrv/article/41/6/bnaa008/5817585>
- 2008 Endocrine Society Diagnosis of Cushing's Syndrome Guidelines:
<https://academic.oup.com/jcem/article/93/5/1526/2598096>
- 2022 Cushing's Disease Consensus Guideline Update:
- <https://pubmed.ncbi.nlm.nih.gov/34687601/>

Pearl: Opposites Principle in Endo

Excess → Failure to suppress
Deficiency → Failure to stimulate

If you suspect hormone *excess*:

- **Check if the hormone fails to fall when it should be low**
 - loss of normal suppression
- **Test whether it can be suppressed** with a suppression test
 - dexamethasone suppression for cortisol, glucose load for GH

If you suspect hormone *deficiency*:

- **Check if the hormone fails to rise when it should be high**
- **Test whether it can be stimulated**
 - Eg ACTH stim/cosyntropin for cortisol

Addison's Disease - Primary Adrenal Insufficiency (AI)

Clinical Presentation

Weakness, lethargy, GI symptoms – not very specific

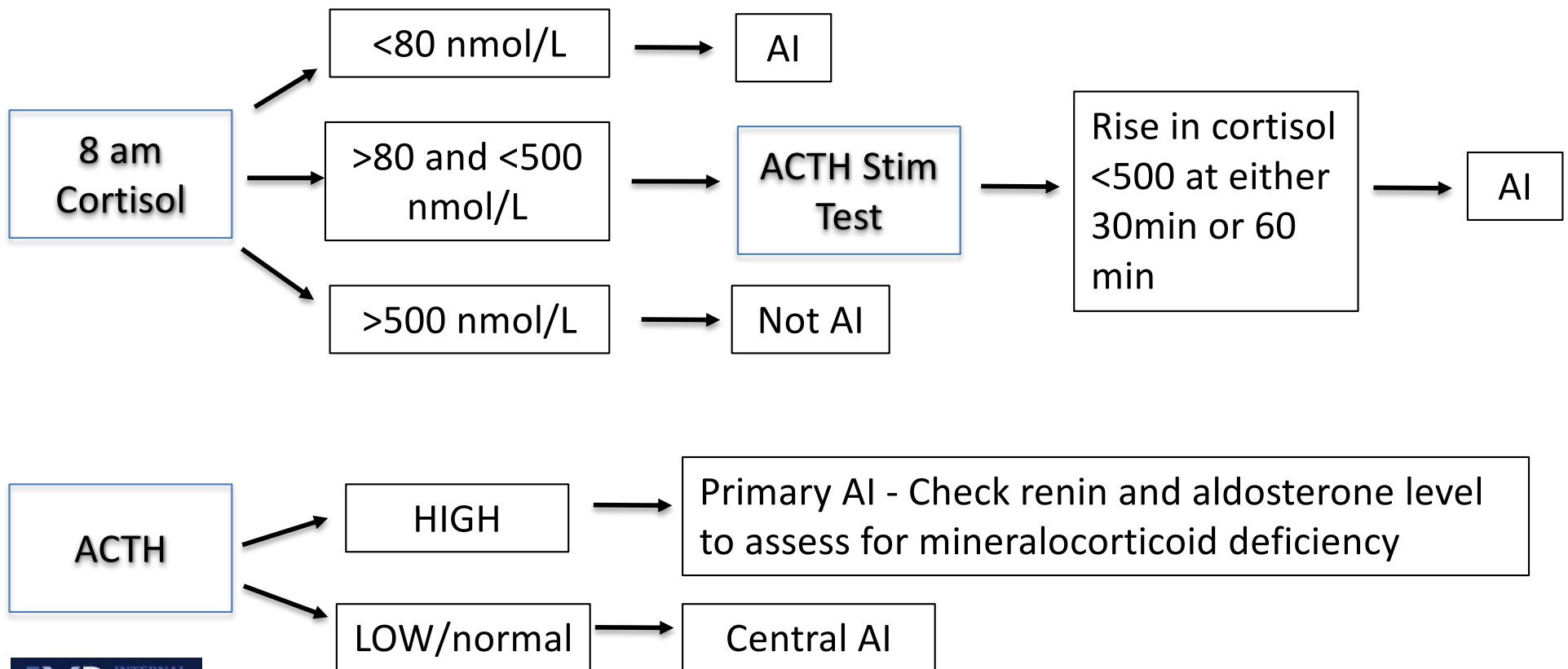
Hyponatremia, hyperkalemia

Orthostatic hypotension, hypoglycemia, uremia

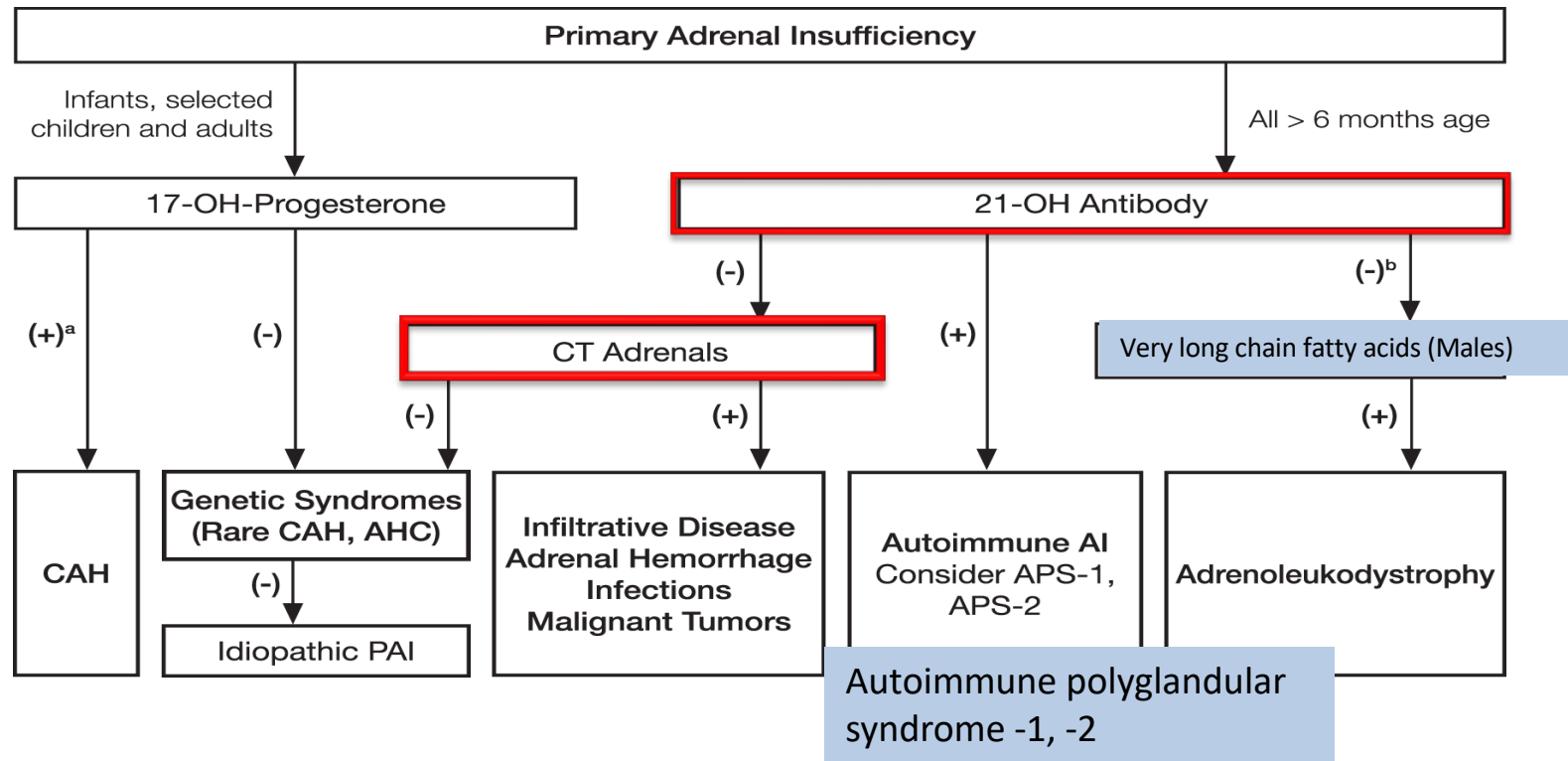
Hyperpigmentation – 1° AI only

- Gingival, Skin furrows, Palmar creases
- Bronzed skin yet areas of vitiligo

Adrenal Insufficiency - Diagnosis



Adrenal Insufficiency - Diagnosis



JCEM 2016. Endocrine Society Guideline: Diagnosis and Treatment of Primary AI

Management of Adrenal Crisis

- Hydrocortisone 100mg IV load followed by 50mg IV Q6H
- IV hydration
- Identify and treat precipitating cause
- Educate patients on sick day rules
 - (eg) Double dose of hydrocortisone when feeling unwell, go to ER if unable to tolerate po
 - advise MedicAlert bracelet

Cushing's Syndrome - Hypercortisolemia

3 Screening Tests – 2 out of 3 needed to establish CS

8mg DST has poor test characteristics – no longer recommended

1) 1mg dex suppression test DST

- A post-dexamethasone cortisol level (<50 nmol/L) is considered “normal” because it suppresses appropriately and **excludes cortisol excess** in most patients.
- Cortisol levels (>140 nmol/L) confirms hypercortisolemia.

2) 24hr urine free cortisol (UFC)

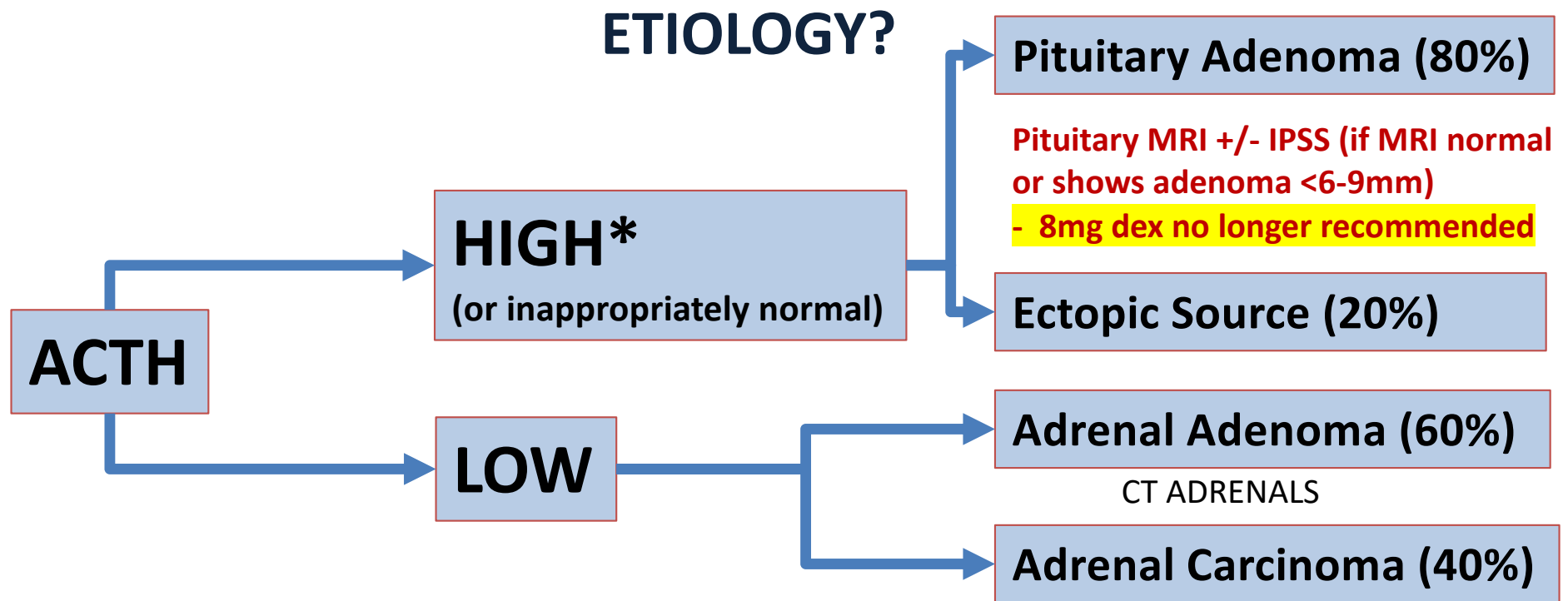
- Positive if abnormal* on 2 separate collections

*Abnormal is any value above the upper limit of lab's reference range

3) Late-night salivary cortisol

- Positive if abnormal* on 2 separate collections

Cushing's Syndrome - Hypercortisolemia



IPSS = Inferior petrosal sinus sampling

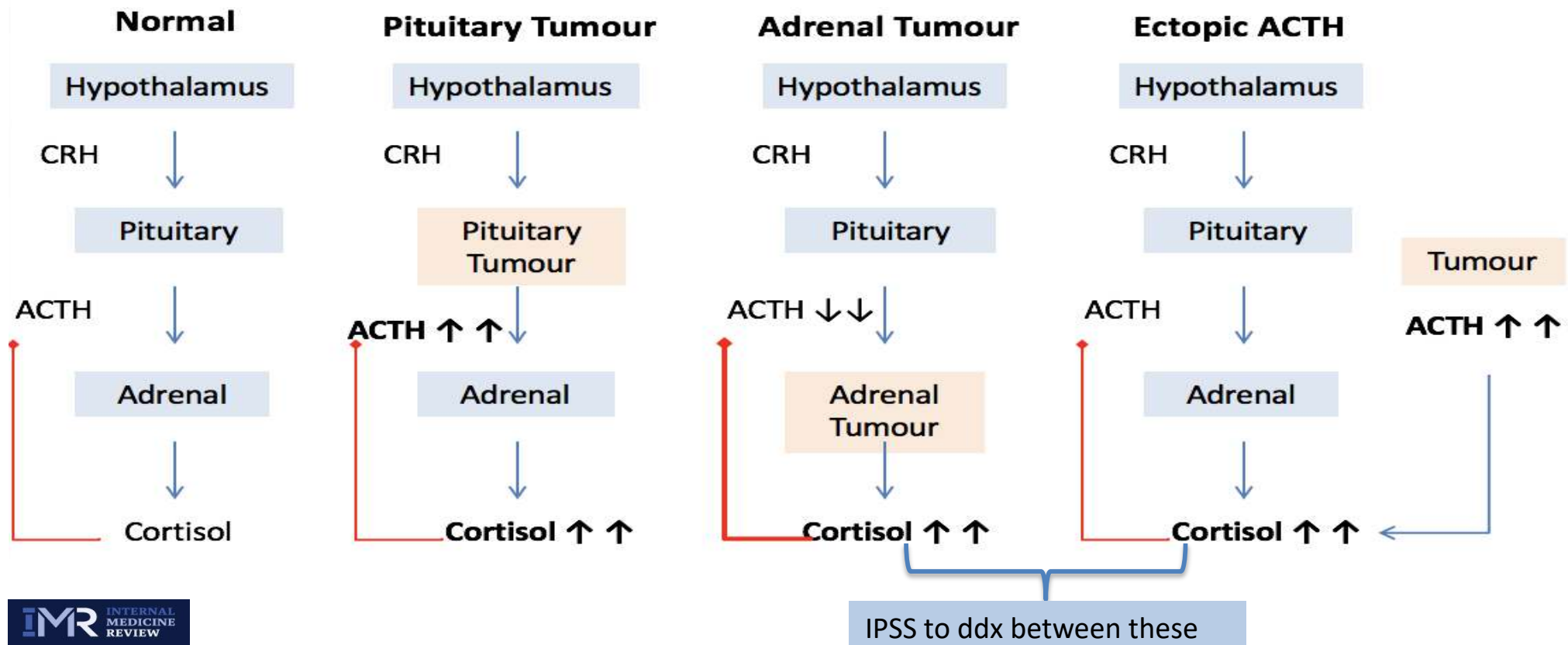
Lancet 2022. Consensus on diagnosis & management of Cushing's disease: a guideline update.

Cushing's Syndrome - IPSS

Inferior Petrosal Sinus Sampling (IPSS):

- Gold standard for differentiating pituitary (CD) vs. ectopic ACTH secretion (usually paraneoplastic)
- Measures ACTH level both at the inferior petrosal sinus (vascular drainage system of pituitary gland) and peripheral venous system
 - Uses a central-to-peripheral ACTH ratio to distinguish pituitary vs. ectopic source
- Invasive, only offered at specialized centres
- Once identify lateralization with IPSS → NEXT STEP IS SURGERY

Cushing's Summary Slide



Hypertension Canada

Conn's Syndrome - Hyperaldosteronism

1) Screen for hyperaldosteronism in hypertensive patients with:

- refractory hypertension (high BP despite ≥ 3 drugs)
- spontaneous hypokalemia ($K < 3.5$) or severe diuretic-induced hypokalemia ($K < 3$)
- adrenal incidentaloma

Screen with Plasma Aldosterone to Renin Ratio (ARR) – check units!

Off: **MRA > ACE/ARB >> BB/DHP-CCB**, clonidine, methyldopa

Can be on: hydralazine, verapamil, doxazosin

2) Confirmatory Tests (remember the principles):

- Saline suppression, oral salt load, captopril suppression

3) CT Abdomen

4) Adrenal Vein Sampling

Unilateral -> surgery

Bilateral or Unilateral w/ non-surgical patient -> MRA

ARR testing pearls:

False positives	Beta-blockers, alpha-2-agonists, oral contraceptives	Stop >4 weeks
False negatives	MRAs, K ⁺ sparing diuretics	Stop >4 weeks
	RASi, thiazide/loop diuretics, dihydropyridine CCBs	Stop >2 weeks

Summary:

Fix: Hypokalemia

Avoid: MRAs, K⁺ sparing diuretics, thiazide/loop diuretics, RASi, beta blockers, OCP (false (+) if using direct renin conc.), NSAIDS

Can use: verapamil/diltiazem, hydralazine, alpha blockers (-zosin)

Hypertension Canada - Pheochromocytoma

- Who to Screen
 - Paroxysmal, unexplained, labile, and/or severe ($\geq 180/110$) sustained HTN refractory to usual therapy
 - HTN + symptoms of PCC (palpitations, headaches, sweating, panic attacks)
 - HTN triggered by beta-blockers, MAO-Is, changes in abdominal pressure, surgery, anesthesia, micturition
 - Adrenal incidentaloma
 - Genetic condition – such as Von-Hippel-Lindau, MEN 2A or 2B, neurofibromatosis type 1
- How to Screen
 - A) Biochemical screening test first
 - **24hr urine total metanephrines and catecholamines (& Cr to ensure adequate collection)**
 - **OR plasma free metanephrines and free normetanephrines**
 - NOT urinary VMA
 - B) Once Biochemical screen confirmed:
 - Confirm adrenal lesion with MR abdomen >> preferred over CT abdomen with delayed contrast washout

Pheochromocytoma

Clinic Management

- Alpha blockade: Phenoxybenzamine or Doxazosin
- Beta blockade ONLY AFTER high dose alpha blockade
- Avoid surgery until at least two weeks of adequate alpha blockade, with liberal salt and fluid intake

Perioperative

- Phentolamine IV if necessary
- Practical recommendations: Pre-op Anesthesia Consult (they need to have bags of vasopressors ready to go as soon as adrenal gland manipulated!), ICU bed postop

Postoperative

- Watch for hypotension, hypoglycemia
- Consider familial syndromes [MEN]

Adrenal Incidentaloma – Structural Evaluation

Consideration #1:

Is it malignant?

- Size >4cm
- Hounsfield Units >10 (>20)
- <50% delayed contrast washout
- Calcifications, extension, adenopathy
- History of malignancy

Consideration #2:

Is it functional?

- ALL:
 - Screen for hypercortisolism or MACS **with 1mg DST – most sensitive test**
 - Screen for pheochromocytoma with 24hr urine or plasma metanephrines
- Screen for hyperaldosteronism (by ARR) **only if HTN +/- Hypokalemia**

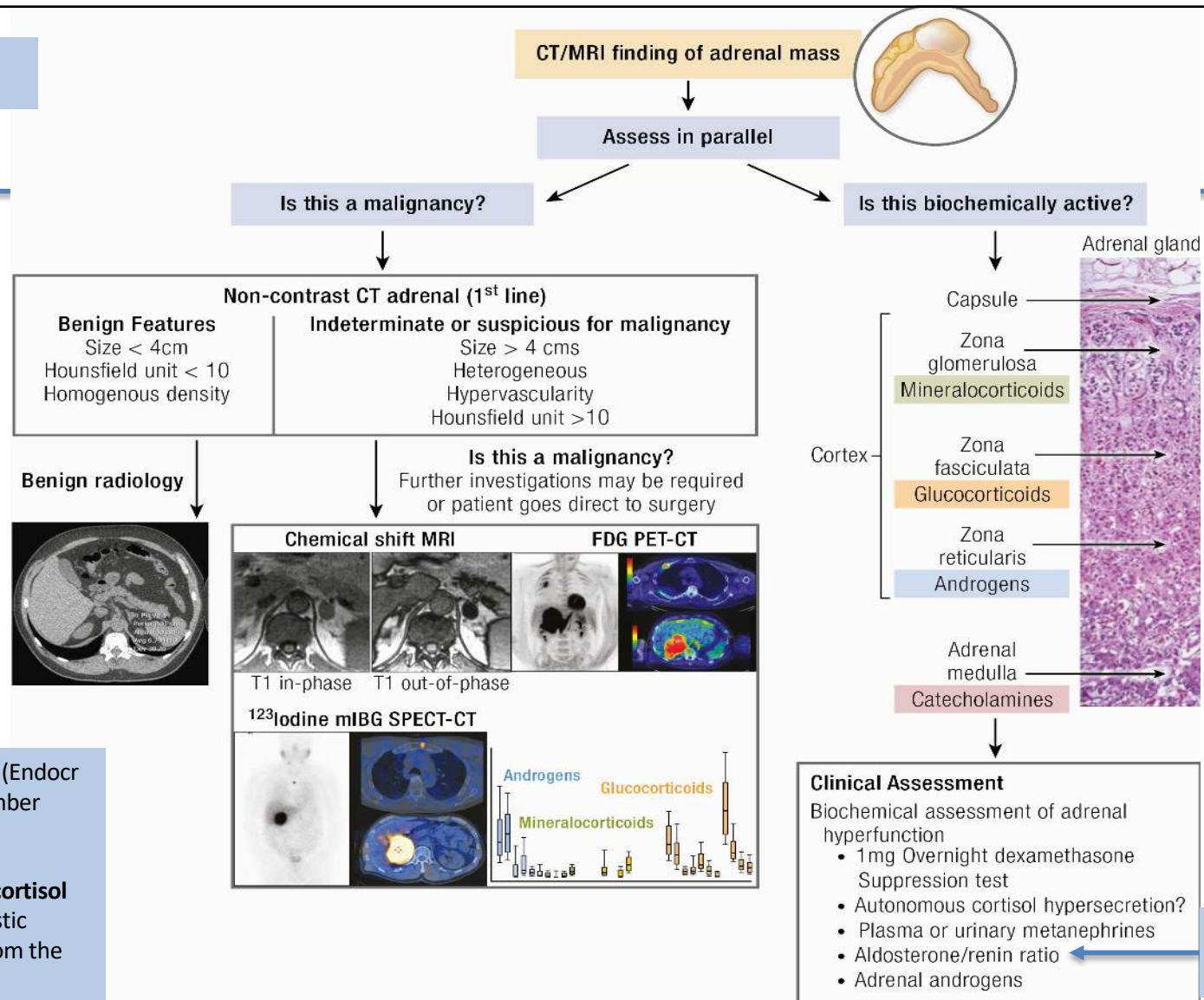
Mild Autonomous Cortisol Secretion (MACS)

- **Mild autonomous cortisol hypersecretion (MACS)**

Incidentalomas produce enough cortisol to result in an abnormal 1mg DST but screen normal on 24hr UFC and late night salivary cortisol testing.

- associated with increased hypertension, hyperglycemia, obesity, dyslipidemia and vertebral fractures
- rare progression to overt Cushing's <1%)
- Treat associated co-morbidities and consider adrenalectomy depending on severity of comorbidities and patient's preference
- **Use 1mg DST as having the highest sensitivity for screening for MACS in all cases of adrenal incidentaloma; with low ACTH +/- low DHEAS as markers of support.**
- *note this differs from CS screening as any of 1mg DST, 24hr UFC, or late night salivary cortisol can be used for CS*

Summary



NEW Adrenal Incidentaloma (Endocr Rev, Volume 41, Issue 6, December 2020, Pages 775–820)

Specifies 1mg DST over other cortisol tests; other portions of diagnostic algorithm have not changed from the previous 2010 JCEM article

MCQ 4 - 2026

A 67 year-old man is referred for low libido. His PMHx is significant for hypertension. His AM testosterone is 7.2 (low) and 6.5 (low) on 2 different mornings. LH and FSH are low. Prolactin is elevated 200. MRI shows a 1.2cm pituitary macroadenoma. What would be the next best step?

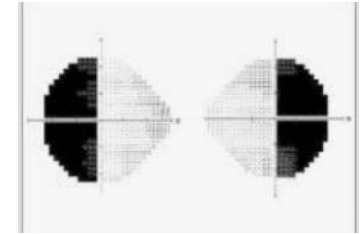
- a. Start testosterone replacement
- b. Resection of pituitary adenoma
- c. Start a dopamine agonist
- d. Testicular ultrasound

Pituitary Tumours

3 Presentations:

1. Mass effect:

- History: headaches, visual changes
- Exam: Cranial nerve exam, Visual fields, MRI
- Bitemporal hemianopsia due to mass effect on optic chiasm



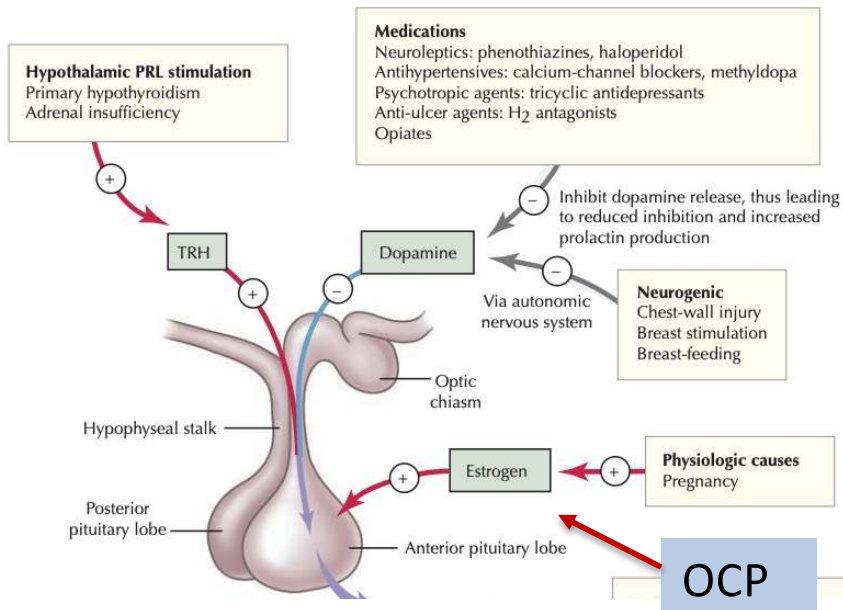
2. Hyperfunction

- Prolactinoma ($\uparrow$ Prolactin), acromegaly ($\uparrow$ GH), Cushing's ($\uparrow$ ACTH), TSHoma ($\uparrow$ TSH)

3. Hypofunction or pituitary deficiency:

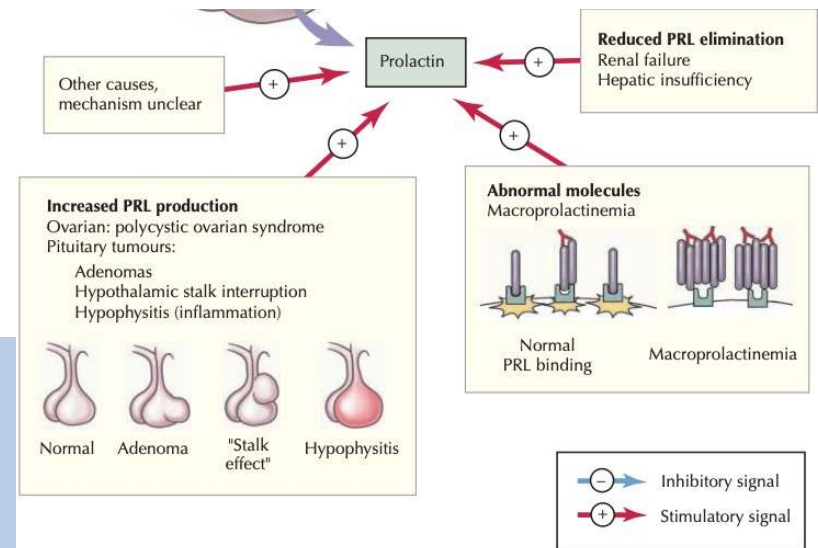
- Lost in the order: “Go Look For The Adenoma, Please”
- GH, then LH and FSH, TSH, ACTH, Prolactin

Hyperprolactinemia



Macroprolactin: type of prolactin multimer which is biologically inactive. It can be in the serum in very high levels and cause no symptoms. It only sticks around because its clearance is slow.

Hook effect: lab peculiarity where very high levels of prolactin are read as low levels. If you dilute the sample, you can get a more accurate reading. Think of this if there are symptoms but PRL not that high.



Disruption of the stalk -> decreased dopamine; Prolactin rarely >100

Lactotroph adenoma -> if ~1cm in size, prolactin ~100. Usually managed medically
 -> **Cabergoline:** more effective, better tolerated, not well studied in pregnancy
 -> **Bromocriptine:** older, more side effects, cheaper
 Side effects: nausea, nasal stuffiness, headache
 -> Surgery if medication resistant/visual compromise/mass effect/hemorrhage

BONUS

Read on own

Acromegaly

Syndrome of growth hormone excess

- Examination: macroglossia, increased ring shoe/collar/glove size, OSA, coarsened facial features, gap between incisors, carpal tunnel, osteoarthritis, type 2 diabetes, visual field defects, cranial neuropathies
 - “spade like hands and feet”
- Screening Test: IGF-1 level
- Diagnostic Test (remember principles): 75 g glucose suppression
- Treatment: Surgical



Andre the Giant
The Princess Bride

Indications for Pituitary Surgery

Optic nerve compression or progressively enlarging tumor with impending compression of the optic chiasm

Hyperfunctioning tumours:
Acromegaly, Cushing's, TSHoma

*Prolactinoma: 1st line Rx is dopamine agonist
(usually shrinks the tumour and no need for surgery)

Pituitary apoplexy - presents with sudden headache, visual disturbances, and acute hypopituitarism due to *hemorrhage or infarction* (most commonly of a pituitary adenoma, but hemorrhage can occur with structurally normal pituitary (i.e. Sheehan syndrome in childbirth)

Management includes corticosteroids, supportive care, and possible surgery for decompression if neurological deficits persist.

Diabetes Insipidus

- Absence of ADH leads to inability to concentrate urine
- Presentation is polydipsia and polyuria, ***usually post-neurosurgery***
 - **Remember that the other possibility is benign dumping fluids from the OR!**
 - Check urine osm: If urine osm > serum osm **it is not DI.**
- Diagnosis:
 - Hypernatremia, hyperosmolar plasma, inappropriately dilute urine
 - 24 hour urine to confirm polyuria
 - Water deprivation test
- Usually presents in post-sellar surgical patients
 - Can be triphasic (or have some or none of):
 1. Transient Diabetes insipidus – Pituitary stunning post-operatively
 2. SIADH – release of stored ADH
 3. Permanent Diabetes insipidus – loss of posterior pituitary function

MCQ 4 - 2026

A 67 year-old man is referred for low libido. His PMHx is significant for hypertension. His AM testosterone is 7.2 (low) and 6.5 (low) on 2 different mornings. LH and FSH are low. Prolactin is elevated 200. MRI shows a 1.2cm pituitary macroadenoma. What would be the next best step?

- a. Start testosterone replacement
- b. Resection of pituitary adenoma
- c. **Start a dopamine agonist**
- d. Testicular ultrasound

The elevated prolactin inhibits the secretion of gonadotropins (LH and FSH), which in turn leads to secondary hypogonadism (low testosterone).

Dopamine agonists (e.g., cabergoline or bromocriptine) are the first-line treatment for prolactin-secreting pituitary adenomas (prolactinomas). These medications help reduce prolactin levels, shrink the tumor, and restore normal gonadotropin and testosterone production.

9. Lipids

Overarching Guidelines: 2021 CCS Guidelines for Management of Dyslipidemia and Prevention of Cardiovascular Disease

[https://www.onlinecjc.ca/article/S0828-282X\(21\)00165-3/fulltext](https://www.onlinecjc.ca/article/S0828-282X(21)00165-3/fulltext)

REVIEW THE LAST PAGE OF THE CCS LIPID GUIDELINE **PDF** that comes after all the references (see below)!

BONUS
Read on own

1150

Canadian Journal of Cardiology
Volume 37 2021



Canadian Cardiovascular Society
Leadership • Change • Connect

AT A GLANCE: 2021 CCS Guideline for the Management of Dyslipidemia in Adults

1. Who to screen with fasting or non-fasting TC, TG, HDL-C, calculated LDL-C and non-HDL-C with ApoB when appropriate and Lp(a) once:

1. Men ≥ 40 yrs old; Women ≥ 40 yrs old or postmenopausal; at younger age in Indigenous and South Asian individuals
2. At any age in patients with:
 - a. Clinical ASCVD
 - b. Evidence of preclinical ASCVD (e.g., CACS or carotid ultrasound abnormalities)

4. Factors not in FRS suggesting that calculated risk may be underestimated:

1. From RCTs:
 - a. JUPITER: CRP ≥ 3.0 mg/L
 - b. HOPE-3: Waist:hip ratio ≥ 0.85 (women) or ≥ 0.90 (men); IFG/IGT, (pre-diabetes, metabolic syndrome)
 - c. ASCOT: in hypertensives, LVH/other EKG abnormalities
2. From epidemiology (consider ethnicity and factors g – p Step 1)

CCS Lipid Guidelines: **Who to Screen**

- Women ≥ 40 years of age (or postmenopausal)
- Men ≥ 40 years of age
- Consider earlier in ethnic groups at increased risk such as *South Asians* or *Indigenous* individuals
- In women with pregnancy-related complication (GDM, HTN, Pre-term birth, Low BW), **screen with complete lipid profile in late postpartum period** (incr. risk of premature CVD and stroke 10-15yr post-delivery)
- **Counsel all women with pregnancy related complication for incr. lifetime risk of ASCVD** and reinforce healthy lifestyle behaviours.
- To assist in lipid-lowering pharmacotherapy decisions in this specific patient population, **use CV age over 10yr risk calculators**
 - **CV age calculator: myhealthcheckup.com**

CCS New Lipid Guidelines: **Who** to Screen

- **ALL** patients with any of the following conditions *regardless* of age

Clinical ASCVD	Family Hx of Dyslipidemia
Evidence of preclinical ASCVD (CACs or carotid U/S abn)	CKD (eGFR <60, ACR >3)
Abdominal aortic aneurysm (AAA)	Obesity (BMI > 30kg/m ²)
Diabetes	Inflammatory diseases (RA, SLE, PsA, AS, IBD)
Hypertension	HIV infection
Current Smoker	Erectile dysfunction
Stigmata of dyslipidemia	COPD
Family Hx of premature CVD in 1 st degree relative	Hypertensive disorder of pregnancy

BONUS

Read on own

Simplified CCS FH Diagnostic Criteria

LDL Cholesterol > 5 mmol/L



Rule out secondary causes

- Obstructive liver disease
- Hypothyroidism
- Nephrotic syndrome
- Anorexia



Positive Family History
First degree relative with
LDL cholesterol > 5 mmol/L
OR
Early coronary heart disease
(<65 years in women, <55 years in men)
OR
Physical findings in patient



Clinical diagnosis of FH



Xanthelasma

**Arcus
Corneus**

Xanthomas

CCS Position Statement on **Familial Hypercholesterolemia** – When FH is suspected: Diagnosis and Treatment Flow
CJC 2014

CCS New Lipid Guidelines: **HOW** to Screen

- History, physical
- Lipid profile **Non-fasting** acceptable as long as TG ≤ 4.5 mmol/L (otherwise, do fasting)
 - Tchol, TG, HDL-c, LDL-c, non-HDL-c
 - Lipoprotein (a) “once”
 - CCS: If Lp(a) > 100 nmol/L, needs earlier and more intensive behavioural modification +/- statin
- Fasting glucose, A1C
- eGFR
- OPTIONAL: apolipoprotein(B), Urine ACR if eGFR < 60 or diabetes or hypertension

RECOMMENDATION

4. We recommend that for any patient with triglycerides > 1.5 mmol/L, non-HDL-C or ApoB be used instead of LDL-C as the preferred lipid parameter for screening (Strong Recommendation, High-Quality Evidence).

Non-HDL: an alternative to LDL



CCS Lipid Guidelines: Who Needs Treatment?

All of these are "statin indicated conditions"

1. LDL >5.0 mmol/L or apoB $\geq$ 1.45 g/L or non-HDL-c $\geq$ 5.8 mmol/L
Or Familial hypercholesterolemia, or genetic dyslipidemia

2. Diabetes if:

- Age $\geq$ 40y
- Age $\geq$ 30y and duration of diabetes over 15yrs
- Microvascular disease

3. CKD if:

- Age $\geq$ 50y and eGFR <60 ml/min/1.73 m² or ACR $\geq$ 3 mg/mmol

4. Atherosclerotic CVD

- MI or ACS, stable angina, documented CAD on angiography
- CVA/TIA/Carotid atherosclerosis
- PAD, Claudication or ABI <0.9
- AAA>3cm or previous aortic aneurysm surgery

- PRIMARY Prevention – Use CV Risk calculator, see future slides.

STATIN Benefits per CCS 2021:

- 20-40% ↓ in LDL for starting dose of statin; additional 6% ↓ for each doubling of dose
- 22% RR of MACE ↓ for each 1 mmol/L reduction in LDL-c

Note KDIGO 2024 CKD guidelines :
>50 years + CKD 1-2: statin.
CKD 3-5: statin or statin/ezetimibe.
(in line w CCS)

Counsel on health/behaviour modification ... AND THEN START STATIN!

CCS Lipid Guidelines: What if **NO** statin indicated condition? **USE FRAMINGHAM RISK**

Statin therapy NOT recommended for most low risk individuals



- Smoking Cessation
- Healthy Diet
- >150 min a week of mod-vigorous intensity aerobic physical activity

Exceptions:

- a) LDL-C ≥ 5.0 mmol/L or ApoB ≥ 1.45 g/L or non-HDL-C ≥ 5.8 mmol/L
- a) FRS 5-9% w/ LDL-C ≥ 3.5 mmol/L or non-HDL-C ≥ 4.2 mmol/L or Apo B ≥ 1.05 g/L particularly with CV risk modifiers
 - Family history
 - Lp(a) ≥ 50 mg/dL (≥ 100 nmol/L)
 - Coronary artery calcium >0 AU

INT (10-19.9%) FRS/ HIGH FRS (>20%):

- Initiate statin
- If on max dose and LDL >2.0 mmol/L, apo B > 0.8 g/L or non-HDL-C >2.6 -> add ezetimibe

If Low FRS <10% but if elevated Lp(a) >100nmol/L they get a statin!



LDL-c Therapies – PCSK-9 inhibitors

Examples: Alirocumab, Evolocumab

Mechanism of Action:

PCSK-9 is a protein which tags LDL-R for recycling. Inhibition causes decreased degradation of LDL-R, and therefore more hepatic uptake of LDL

- Effect: ~50-60% ↓ in LDL-R, regardless of LDL or background therapy
- Current indications:
 - Heterozygous Familial Hypercholesterolemia
 - Clinical atherosclerotic cardiovascular disease
 - Homozygous Familial Hypercholesterolemia

High PCSK9i benefit:

- Recent ACS <1 yr
- ASCVD + any of the following:
 - DM/metabolic syndrome
 - Polyvascular disease (vascular disease >2 arterial beds)
 - Symptomatic PAD
 - Recurrent MI
 - MI in past 2 yrs
 - Previous CABG
 - LDL $\geq$ 2.6 mmol/L or heterozygous HH
 - Lipoprotein (a) > 120 nmol/l

CCS Lipid Guidelines: Treatment Summary

Condition (statin indicated)	LDL goal	apoB goal	non- HDL-C	First line* Rx	Add-on #1 **	Add on #2 **
LDL>5	<2.5	<0.85	<3.2	STATIN for all conditions	Ezetimibe	PCSK9i
DM or CKD	<2.0	<0.80	<2.6		Ezetimibe	Bile acid sequestrant
Intermed FRS or high FRS	<2.0	<0.80	<2.6		Ezetimibe	Bile acid sequestrant
ASCVD or AAA (2° prevention)	<1.8	<0.7	<2.4		EZETIMIBE: LDL 1.8-2.2 <i>or</i> apoB 0.7-0.8 <i>or</i> non HDL-C 2.4-2.9 PCKS9i: LDL >2.2 <i>or</i> apoB >0.8 <i>or</i> non-HDL-C >2.9 <i>or</i> PCKS9i benefit pt*	
*All rows = non-pharmacologic 1 st line ... Health Behaviour Modification! • Smoking cessation • 150min/week mod vigorous activity • Dietary modifications ** Discuss cost, access, side effects at every juncture						



What about Triglycerides Targets?

- **Fasting TG ≥ 1.5 -5.6 AND**
 - ASCVD on maximally tolerated statin (with LDL >1)OR
 - DM on max tolerated statin with ≥ 1 CV risk factor
 - Men ≥ 55 y and Women ≥ 65 y
 - Smoker currently or within 3 mos
 - HTN
 - HDL <1.04 men or <1.3 women
 - hsCRP >3
 - eGFR 30-60 ml/min
 - Retinopathy
 - Albuminuria
 - ABI <0.9
- What? **Icosapent Ethyl 2g BID (VASCEPA)**
 - \$3500/ year

10. Care of Transgender Patients

Terminology in Canada

- Be sensitive, **ask patient name and pronouns**
- **Must provide contraceptive counseling to transmen having receptive intercourse (testosterone is androgenizing to fetus; can still get pregnant on testosterone)**
- **MUST OFFER fertility referral at initial appointment and document that discussion of potential reduced fertility [send referral if they are interested in banking eggs/sperm]**
- **Counsel smoking cessation; smoking is NOT an absolute contraindication to estrogen**
- **Must be on GAHT if gonadectomy (or will get osteoporosis)**

Endocrine Society Guidelines 2017:

Transwomen/femme patients have increased risk of VTE and Br Cancer compared to Cis-men
Transmen/masc patients have increased erythrocytosis, metabolic syndrome, OSA and often have higher rates of PCOS prior to gender affirming hormone therapy (GAHT)

Long-term studies of CV risk ongoing

GOOD LUCK!

- Bonus Slide Content:
 - Reproductive endocrinology
 - Thyroid Nodules
 - Obesity
 - potential oral scenario
- Bonus Questions for self study



11. Reproductive Endocrinology

Resources:

- 2015 Endocrine Society Guidelines on Treatment of Symptoms of the Menopause
<https://academic.oup.com/jcem/article/100/11/3975/2836060>
- 2013 Endocrine Society Guidelines on PCOS
<https://academic.oup.com/jcem/article/98/12/4565/2833703>
- 2018 Endocrine Society Guidelines on Hirsutism
<https://academic.oup.com/jcem/article/103/4/1233/4924418>
- 2015 ESHRE Guidelines on POI
<https://academic.oup.com/humrep/article/31/5/926/1749616>
- 2018 Endocrine Society Guidelines on Testosterone Therapy for Hypogonadism
<https://www.endocrine.org/clinical-practice-guidelines/testosterone-therapy>

BONUS

Read on own

BONUS

Read on own

Menopause

Definition: Cessation of menses for 12 mo in a previously cycling woman and reflecting complete or nearly complete permanent cessation of ovarian function and fertility.

Symptoms:

- Vasomotor symptoms - hot flashes, night sweats
- Genitourinary symptoms - vaginal dryness, dyspareunia
- mood changes
- declining bone mass/fractures
- sleep disruption

**Treatment of Symptoms of the Menopause:
An Endocrine Society Clinical Practice
Guideline**

J Clin Endocrinol Metab, Volume 100, Issue 11,
1 November 2015, Pages 3975–4011,
<https://doi.org/10.1210/jc.2015-2236>

Indications for Hormone Replacement Therapy:

- symptomatic (VMS) and < 60 years old or <10 years since start of menopause
- Know contraindications to HRT:
 - MI, stroke/TIA, PE/VTE
 - breast or endometrial cancer
 - unexplained vaginal bleeding
 - active liver disease
- Caution – hypertriglyceridemia, active gallbladder disease, migraine with aura, increased risk of CVD or breast cancer, diabetes, smoking, HTN

Evaluate CV risk + breast cancer risk

If uterus → estrogen + progesterone; if no uterus → estrogen alone reasonable

Non-hormonal treatments: SSRI/SNRI, gabapentin, oxybutynin, CBT

Fezolinetant: NK3R antagonist blocks NKB binding to the KNDY neuron to reduce vasomotor symptoms

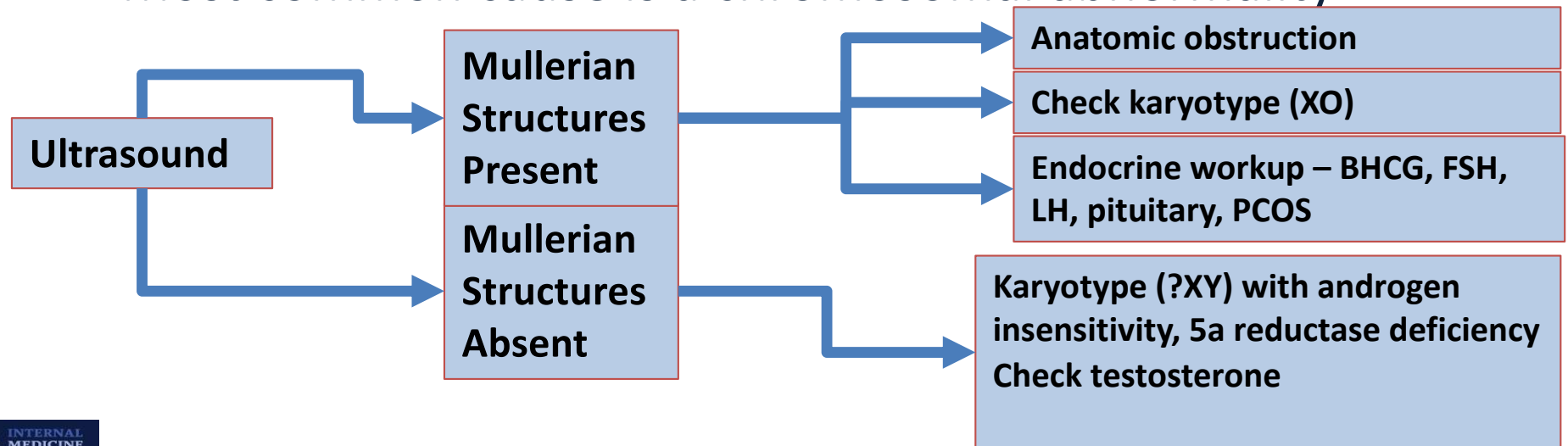
BONUS

Read on own

Amenorrhea - Primary

Definition:

- Absent menses at age 16
- Absent sexual characteristics at age 13
- Most common cause is a chromosomal abnormality



BONUS

Read on own

Amenorrhea - Secondary

1. Hypothalamic – Functional hypothalamic amenorrhea
 - Low LH, FSH and low estradiol -> high performance athlete, eating disorder, high stress
2. Pituitary – hyperprolactinemia from prolactinoma or other sellar mass causing stalk effect, infiltration (hemochromatosis), Sheehan's, Cushing's
3. Thyroid - hyperthyroidism
4. Ovarian – PCOS (#1), POI
 - *PCOS is insulin resistance at the level of the ovary, this is why metformin may help regularize cycles*
5. Pregnancy

Definition: No cycles for >3 months in those with regular periods, or >6 months in those with irregular periods

Other structural considerations: bicornuate uterus, endometriosis, Asherman's syndrome

BONUS

Read on own

Ovarian Disorders

- PCOS
 - Diagnosis – 2/3 of the Rotterdam criteria:
 - **Menstrual irregularity**
 - **Biochemical or clinical hyperandrogenism (hirsutism, acne, androgenic alopecia)**
 - **Ultrasound findings of polycystic ovaries**
 - Exam: BMI, Ferriman-Gallwey score, alopecia, acne, acanthosis nigricans
 - Investigations: Prolactin, TSH, 17OHP, testosterone, DHEAS, LH, FSH, estradiol, US pelvis, fasting BG, **2hr OGTT**, lipid profile, sleep study
 - Issues: hyperandrogenism, fertility, metabolic health
- Primary Ovarian Insufficiency (loss of ovarian activity at age <40)
 - High FSH, Low Estradiol
 - History: Autoimmune, Radiation, Chemotherapy, Fragile X (FHx intellectual disability/ataxia)
 - Tests: Karyotype (for Turner syndrome), Fragile X, **Anti-adrenal antibodies, Anti-TPO antibodies**
 - Screening: BMD
 - Rx: HRT until age of usual menopause (52) for CV and bone protection

BONUS

Read on own

PCOS Management Options

	Endometrial Protection (anovulation)	Insulin resistance	Hirsutism	Fertility
Metformin	x	x	(possible benefit but unlikely)	(as an adjuvant to clomiphene to prevent ovarian hyperstimulation if undergoing IVF)
OCP (first-line)	x		x	
Spironolactone			x	
Clomiphene, letrozole				X (ovulation induction)
Weight loss	x	x	x	x

BONUS

Read on own

Hyperandrogenism in Women

- **History and Physical**
 - Acne, voice changes, mood
 - hirsutism (do random total testosterone if abnormal hirsutism score)
 - Clitoromegaly, menstrual abnormalities
 - Hypothyroidism, Liver disease
 - Anabolic steroids?
- **Lab Investigations**
 - Total testosterone/free testosterone
 - DHEA-S; if normal, not an adrenal problem
 - 17-OH Progesterone to assess for non-classical congenital adrenal hyperplasia (NCCAH) in women with high testosterone or high risk for CAH – increased in CAH, **Non-classic CAH (>6mmol/L warrants ACTH stim testing)**
 - Prl, TSH
 - FSH/LH (In PCOS LH>FSH)
 - Image ovaries and adrenals if Testosterone very high (>5 mmol/L)
 - Cushing's screen if clinical features suggestive
- **Virilization** – increased muscle bulk, voice, clitoromegaly *red flags!* → not PCOS → look for tumor-related androgen production e.g. **adrenocortiocarcinoma (ACC)**

Therapy for hirsutism:

- Cosmetic measures (plucking, tweezing, waxing)
- Local hair removal methods (photoepilation, electrolysis)
- Lifestyle changes (if obese or PCOS)
- 1st line Pharmacotherapy: oral contraceptive pill
- 2nd line Pharmacotherapy: anti-androgen (spironolactone) in combo with OCP if no improvement after 6 months of monotherapy
- Anti-androgens not recommended as monotherapy due to teratogenic potential unless a woman is not sexually active, has undergone permanent sterilization, or is using long-acting reversible contraception

Endocrine Society 2018: Evaluation and Treatment of Hirsutism in Premenopausal Women

<https://www.endocrine.org/guidelines-and-clinical-practice/clinical-practice-guidelines/hirsutism>

BONUS

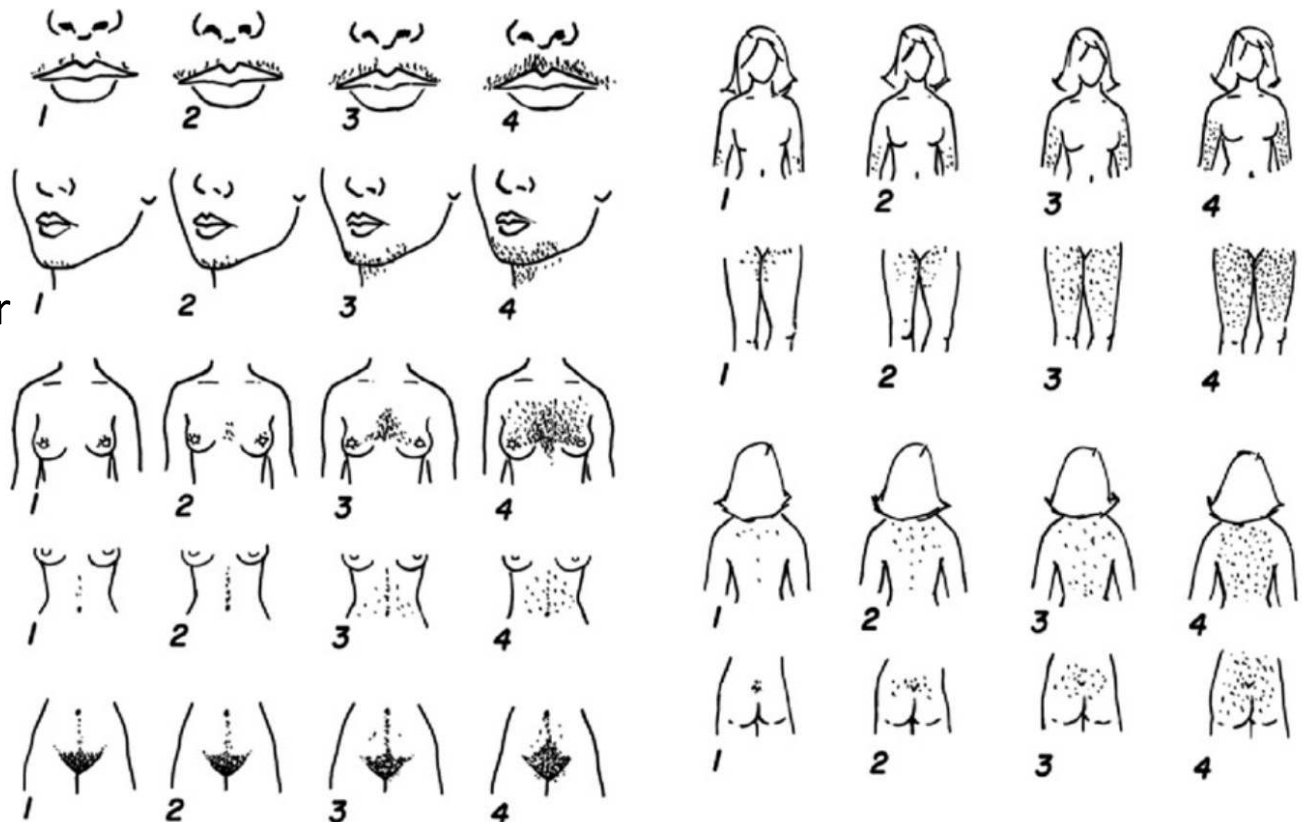
Read on own

Ferriman-Gallwey Score

Generalized Hirsutism ≥ 8

Locally Excessive Growth < 8

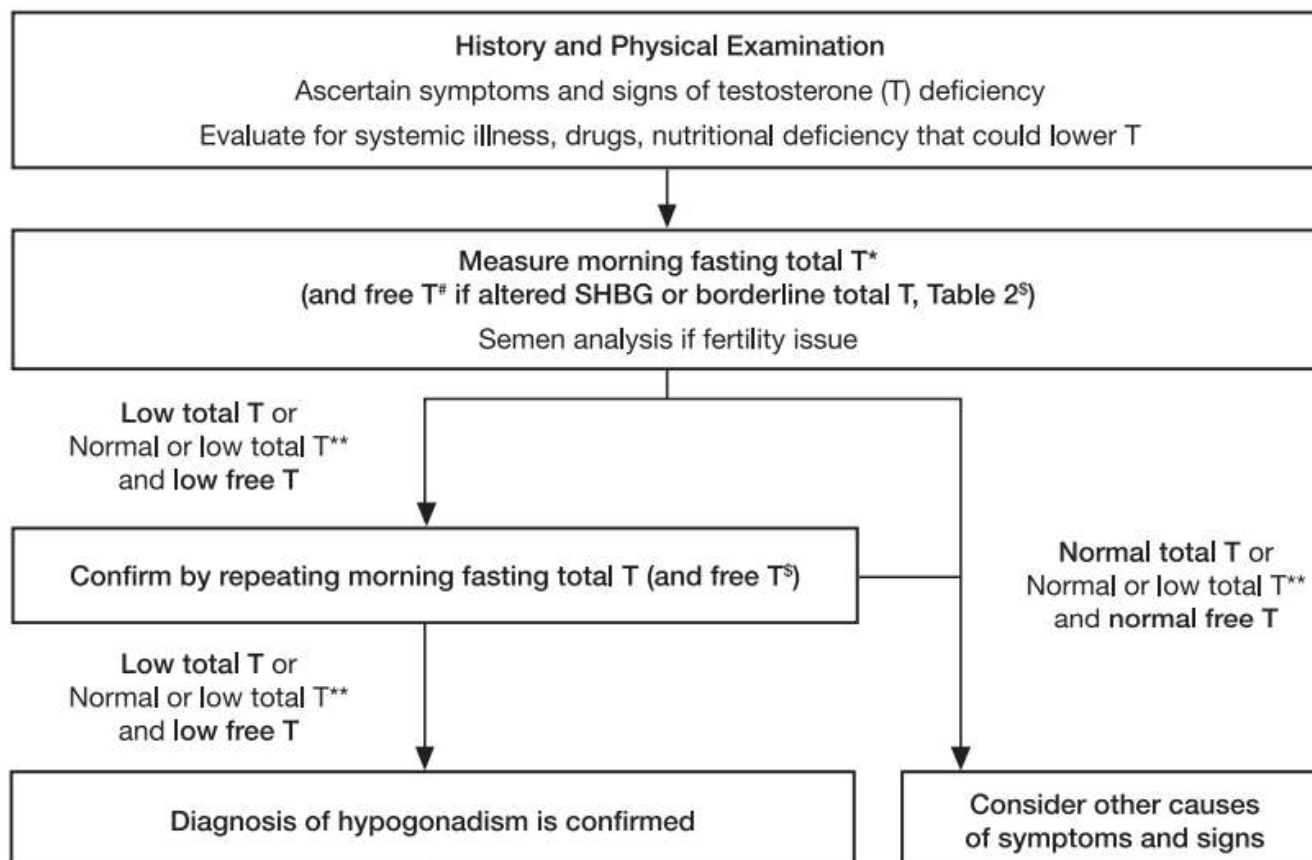
Note: A “normal” score is lower in some Asian populations and higher in some Mediterranean populations



BONUS

Read on own

Male Hypogonadism - Approach



If any conditions that alter SHBG, screen with free testosterone (FT) using equilibrium dialysis assays or calculations that involve TT, SHBG, and albumin (other FT assays are inaccurate)

BONUS

Read on own

Male Hypogonadism: 1° vs 2° Causes

- Primary: High FSH/LH → Klinefelter (XXY – do karyotype), chemotherapy, testicular trauma, mumps, autoimmune, systemic illness
- Secondary: Low FSH/LH → pituitary adenoma, hyperprolactinemia, hemochromatosis, anabolic steroids, obesity, opioids, DM2
- Klinefelter's Syndrome
 - Small firm testes
 - Increased LH, FSH, decreased testosterone
 - Arm span > Height + 5cm
 - Leg length > upper body + 2cm
 - Comorbid: NHL, bronchiectasis, SLE, diabetes, osteopenia, breast cancer
 - Diagnosis: Karyotype

BONUS

Read on own

Male Hypogonadism – Recommendations

- Measure total testosterone
- Therapy
 - Recommended to maintain secondary sex characteristics and to correct symptoms of testosterone deficiency
 - Not recommended if planning fertility in the near term, breast or prostate cancer, prostate nodule/induration, high PSA, high hematocrit, untreated OSA, severe LUTS, uncontrolled heart failure, MI or stroke in the last 6 months, or thrombophilia
 - Caution in older men → discuss risks and benefits
- Monitoring
 - Symptoms, testosterone levels, hematocrit, PSA
 - Several times during the first year then annually

BONUS

Read on own

12. Thyroid Nodule & Cancer

Resources:

- 2015 ATA Guidelines on Thyroid Nodules & Differentiated Thyroid Cancer
- <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4739132/>
- ACR TI-RADS Assessment Categories
- <https://www.acr.org/-/media/ACR/Files/RADS/TI-RADS/TI-RADS-chart.pdf?la=en>

BONUS

Read on own

Thyroid Nodule Work-Up

Differential Diagnosis:

- Multinodular goiter, Toxic nodule, Benign non-functional nodule, Cyst
- Thyroid cancer (Papillary 90%, Follicular, Medullary, Anaplastic)
- Lymphoma
- WORKUP:
 1. History and physical examination
 2. TSH
 - a) Do RAI Uptake/Scan if TSH <0.5 *don't stick a needle in a hot nodule*
 3. Ultrasound *because if it's 3cm but purely cystic, you wouldn't biopsy*
 4. FNA

Thyroid Nodule – Who needs a biopsy?

BONUS

Read on own

Determined based on TIRADS scoring (don't memorize, know the high-risk features):

1. High suspicion sonographic features (TR5): >1cm
2. Moderate suspicion sonographic features (TR4): >1.5cm
3. Mild suspicion sonographic features (TR3): >2.5cm
4. Do not biopsy if TR1-2: Purely cystic nodules

*If there are multiple nodules, treat each nodule individually and make a decision on whether to biopsy that nodule.

BONUS

Read on own

Thyroid Nodule & Thyroid Cancer

- High risk features on US:
 - Hypoechoic
 - Irregular margins
 - **Microcalcifications** *highest specificity feature for thyroid ca on U/S*
 - Taller than wide
 - Extrathyroidal extension
 - Peripheral (rim) calcifications
 - Lymphadenopathy
 - >20% increase in 2 dimensions

Features post-total thyroidectomy of:

+LNs, margins, extrathyroidal extension, distant mets, **needing RAI (high-risk!)**

- Point you to Intermediate-HIGH risk category:

Low-risk = TSH goal of 0.5-2.0mU/L

Intermediate-risk = TSH goal of 0.1-0.5mU/L

High-risk = TSH goal <0.1mU/L

*Note if TSH is suppressed <0.1, and/or the nodule is “hot” do not biopsy it, this is a **LOW risk** feature and warrants investigation for Graves, toxic adenoma, and multi-nodular goitre*

BONUS

Read on own

13. Obesity

Obesity in adults: a clinical practice guideline (2020 + 2022 Pharmacotherapy Update)

<https://www.cmaj.ca/content/192/31/E875>

<https://obesitycanada.ca/guidelines/pharmacotherapy/>

- Focus has shifted from a physical characteristic (BMI) to the science behind obesity and drivers of weight gain.
- **New Definition:** Obesity is a prevalent, complex progressive, and relapsing chronic disease characterized by abnormal or excessive body fat that IMPAIRS HEALTH (functional, medical, psychosocial, metabolic health).
- Cognitive functions in the prefrontal cortex exert executive control on food choices and the decision to eat. The interconnectivity of these neural networks drives eating behaviour and has been shown to be altered in obesity.
- Risk of metabolic disease (T2DM, CV Disease) well known
- BUT: It is estimated that 20% of all cancers can be attributed to obesity, independent of diet.
 - Obesity increases the risk of colon, kidney, esophageal, endometrium, and post-menopausal breast cancer

BONUS

Read on own

Pharmacotherapy – 2025 Update

- Indicated if BMI ≥ 30 or BMI ≥ 27 with adiposity-related complications (T2DM, gallbladder disease, NAFLD, gout)
 - *In conjunction with medical nutrition therapy, physical activity, +/- psychological interventions
- Options include:
 - Semaglutide 2.4 mg weekly (Wegovy)
 - Tirzepatide 5 mg, 10 mg, or 15 mg weekly (Zepbound)
 - Liraglutide 3 mg daily (BMI $\geq 27^*$) (Saxenda)
 - Naltrexone–bupropion 16 mg/180 mg twice daily (Contrave)
 - Orlistat 120 mg PO TID (BMI 28–47*) (Xenical)

BONUS

Read on own

Bariatric Surgery

Referral indications:

BMI 35-40 AND 1 serious comorbidity (DM2, DLD, CAD, HTN, pseudotumor cerebri, obesity hypoventilation syndrome, debilitating OA, NAFLD, OSA, severe GERD)

BMI $\geq$ 40 without comorbidity

Complications:

General surgical risks

Stomal obstruction, gastric perforation, GI bleed, biliary stone disease, hernia, dumping syndrome, **post-prandial hypoglycemia**

Malabsorption: iron, B12, folate, calcium, vitamin ADEK, thiamine

Post-operative Management

Specialized center – multidisciplinary

Monitor and titrate therapies – DM/HTN may be cured

Nutritional deficiencies (no guideline), BMD annually

BONUS

Read on own

Meds Associated with Wt. Gain & Possible Alternatives

Category	Ex of Drugs Causing Weight Gain	Possible Alternatives
Neuroleptics	Olanzapine, quetiapine, risperidone, clozapine, haloperidol	Ziprasidone, lurasidone, aripiprazole
TCAs	Amitriptyline, nortriptyline, doxepin	Will depend on the reason for use
SSRIs	Paroxetine, citalopram	Fluoxetine, sertraline, bupropion, duloxetine, venlafaxine
Other anti-depressants/ mood stabilizers	Mirtazapine, phenelzine, lithium	Another anti-depressant class
Anticonvulsants	Valproate, carbamazepine, gabapentin	Topiramate, lamotrigine
Antihyperglycemic meds	Insulin, sulfonylurea, TZDs	Acarbose, MTF, GLP1RA, SGLT2i, DDP4i
Antihistamines	First-generation anti-histamines	Second-generation anti-histamines
Beta-Blockers	Propranolol	ACEi, ARB, CCBs
Steroid hormones	Contraceptives, glucocorticoids	Other contraceptive devices

Bonus MCQ 1 2026

A 58-year-old man with type 2 diabetes, hypertension, and obesity (BMI 33 kg/m²) presents with progressive fatigue, decreased libido, and reduced morning erections over the past year.

Exam: central obesity, mild loss of body hair, and small but symmetric testes.

Initial labs show: Total testosterone: 8.2 nmol/L (reference 10–30), Free testosterone (calculated): 170 pmol/L (normal 200–600), normal LH and FSH, normal prolactin, normal TSH

Which of the following is the most appropriate next step?

- a) Start testosterone replacement therapy immediately
- b) Repeat morning total testosterone with SHBG and consider pituitary MRI if low again
- c) Order a karyotype to evaluate for Klinefelter syndrome
- d) Refer directly for testicular biopsy

Bonus MCQ 2 2026

A 45-year-old woman is found to have a 0.7cm L incidental adrenal mass on imaging. She does not have a history of hypertension and does not appear Cushingoid. Which tests would you do next?

- A) Only serum metanephrines
- B) Serum metanephrines + 24 hour urine cortisol
- C) Serum metanephrines + dexamethasone suppression test
- D) Serum metanephrines, dexamethasone suppression test, aldosterone-renin-ratio

Bonus MCQ 3 2026

A 28-year-old woman presents with progressive facial and body hair growth over the past 2 years. She reports irregular menses and difficulty conceiving. Her BMI is 32 kg/m². Labs show normal TSH and prolactin, and mildly elevated total testosterone. Ultrasound shows polycystic ovaries. She is primarily concerned about excess hair growth on her chin, upper lip, and lower abdomen.

Which of the following is the most appropriate first-line treatment for her hirsutism in the context of PCOS?

- a) Oral contraceptive pills (estrogen + progestin)
- b) Spironolactone
- c) Finasteride
- d) Metformin

Bonus MCQ 4- 2025

A 43-year-old woman presents with 3 months of palpitations, anxiety, increased bowel movements, and 4 kg weight loss. On examination, BP 128/70, HR 104 and regular, and thyroid feels firm. On eye exam, she has proptosis and conjunctival redness. Her thyroid tests and TRAb are pending. Which of the following statements about thyroid eye disease (TED) is TRUE?

- A) Smoking has no effect on the severity or progression of TED
- B) TED symptoms always improve after treatment of hyperthyroidism
- C) The most common initial symptom of TED is loss of vision
- D) TED can occur in both hyperthyroid and euthyroid patients

Bonus MCQ 5 2025

50 yo F being worked up for secondary HTN and hyperaldosteronism. Which medication is it okay for her to take prior to being tested for hyperaldosteronism?

- a) Spironolactone
- b) Verapamil
- c) Metoprolol
- d) Candesartan

Bonus MCQ 6 2025

50 yo M admitted with colitis has incidental finding of left adrenal adenoma. Which of the following radiologic features would be most in keeping with a malignant lesion?

- a) 9 Hounsfield units
- b) <50% delayed contrast wash out
- c) No calcifications present
- d) 3.3 cm nodule

Bonus MCQ 7 2025

75-year-old male POD 1 transsphenoidal resection of pituitary macroadenoma resection presents with fatigue, mild nausea, excess thirst and increased urine output. His vitals show BP 110/74, HR 64, T 37.2, 98% RA. Urine output for the last two hours: 450cc and 500cc. Blood work shows Na 148 K 4.2 Cr 133. Urine Osm 76. What is the cause of this patient's presentation?

- a) IV fluids from OR
- b) Diabetes insipidus
- c) Adrenal crisis
- d) Hypothyroidism

Bonus MCQ 8 2025

56 yo F known for hypertension on hydrochlorothiazide and perindopril. She presents to your clinic with several months of worsening fatigue. You do routine blood tests which includes a total Ca 3.05, normal phosphate, PTH 6.8 (normal <6.6), and PTHrP normal. What is the most likely cause of the hypercalcemia?

- a) hydrochlorothiazide
- b) hypercalcemia of malignancy
- c) primary hyperparathyroidism
- d) vitamin D excess

Bonus MCQ 9 2025

24 M with fatigue and 15 lbs weight loss in 6 months. His systolic BP drops from 130 to 90s with standing. His skin appears hyperpigmented. Na 129, K 5.3. Cortisol 18, ACTH high. What is your next step?

- a) Abdomen CT
- b) MRI abdomen
- c) PO hydrocortisone
- d) PO hydrocortisone and fludrocortisone

Bonus MCQ 10 2024

23 year old female comes in with a serum prolactin of 50. She's completely asymptomatic. She is not on any medication is known to increase production. Her neurologic examination, physical examination and breast examination are all normal. What do you do next?

- a) Serum macroprolactin
- b) Visual fields
- c) MRI of Sella
- d) Transvaginal ultrasound

Bonus MCQ 11 - 2024

A 65-year-old female with bipolar disorder on valproate and osteoporosis on alendronate, calcium, and vitamin D presents with leg pain after stumbling. She is found on X-ray to have an atypical femoral fracture. Secondary osteoporosis work-up was negative and her vitamin D level normal. Which of the following would be the next best step?

- a) Stop oral bisphosphonate and start yearly IV bisphosphonate
- b) Continue oral bisphosphonate
- c) Stop alendronate and start denosumab
- d) Stop alendronate and start teriparatide

BONUS MCQ 12- 2025

A 68 yo M presents to primary care. He has a history of COPD, still smoking and has had frequent exacerbations requiring hospitalization. A recent CXR done during exacerbation noted new vertebral fracture at T7. The patient has lab-work done, showing Ca 2.17, Alb 38, ALP N, TSH N, eGFR 48, PTH 11.7, 25-OH-Vit D 38 nmol/L

A BMD/DXA Scan is done and FRAX is calculated at 16% 10 year risk of major osteoporotic fracture. Which of the following is appropriate to recommend at this time?

- a) Alendronate 70mg po q weekly
- b) Exercise and resistance training weekly
- c) Vitamin D 1000 IU daily
- d) Denosumab 60mg sc q 6 mos
- e) No treatment for moderate risk FRAX, recommend non-pharm therapy and adequate calcium and Vitamin D
- f) Calcium 1000 mg daily

BONUS MCQ 13- 2026

A 22-year-old woman with type 1 diabetes (diagnosed 4 years ago) is on a multiple daily injection (MDI) regimen with glargine 20 units at bedtime and aspart 7 units with meals. She has recently become more consistent with her bolus insulin use.

Her CGM report shows:

Fasting glucose: 9.4 mmol/L

Pre-lunch: 14.2 mmol/L

Pre-supper: 6.1 mmol/L

Bedtime (3h post-supper): 12.9 mmol/L

Time in range (3.9–10 mmol/L): 52%

What is the most appropriate next step in her management?

- A) Initiate insulin pump therapy
- B) Increase glargine to 24 units at bedtime, maintain current bolus doses
- C) Increase glargine to 22 units at bedtime, Increase aspart to 9 units at breakfast and supper
- D) Initiate carbohydrate counting with a ratio of 1 unit: 6 g carbohydrate at each meal

BONUS MCQ 14 - 2025

A 67 year-old man is referred for cardiovascular risk management. His PMHx is significant for symptomatic PAD. His medications include atorvastatin 80 mg po QHS. Latest lipid panel shows total cholesterol 3.2, HDL 0.95, LDL 1.7, TG 1.2, Apo-B 0.94. Which of the following would you add based on the 2021 CCS Lipid Guidelines?

- a. No medication needs to be added
- b. Icosapent ethyl 2g BID
- c. Ezetimibe 10mg daily
- d. Evolocumab 140mg q2 weeks

Bonus MCQ 15 - 2025

An 85yoF with bony pain is referred for hypercalcemia. Meds: ramipril, atorvastatin, glargine, aspart. A CT scan done shows lytic lesions. sPEP, uPEP, FLC assay and malignant workup is normal. BMD do not have any T-scores <-2.5 at any sites (including radius). A sestamibi scan shows a 1.3 cm parathyroid adenoma. Her nephrologist and ENT determine she is not a candidate for parathyroidectomy due to her frailty and she has also declined surgery. What treatment option is best to medically manage her hypercalcemia?

1. IV fluids + furosemide
2. Calcium restricted diet
3. IV bisphosphonate
4. Cinacalcet
5. Calcitriol

Lab Values
Calcium = 2.74 (H)
Phosphate = 0.90 (N)
Cr = 123 (H)
25OHD = 74 (low-N)
PTH = 35 (H)
Albumin = 40 (N)

BONUS Answers

- 1) B – repeat morning testosterone, ideally along with SHBG to assess bioavailable/free testosterone. If low, central hypogonadism should be evaluated further, including pituitary imaging to rule out mass lesions
- 2) C – All adrenal incidentalomas should have screening for pheochromocytoma with serum metanephrines and for MACS (mild autonomous cortisol secretion with a 1mg dexamethasone suppression test – most sensitive test)
- 3) A – Combined oral contraceptive would be the first line option. You would not prescribe spironolactone alone without reliable contraception due to teratogenicity.
- 4) D – TED can occur even when euthyroid. Smoking cessation worsens it. Symptoms persist even during treatment for hyperthyroidism. Initial TED symptoms are usually eye redness, irritation, and eyelid retraction. Vision loss occurs in more severe cases.
- 5) A – Verapamil is only medication that won't alter her aldosterone testing.
- 6) B – <50% delayed contrast washout is a feature of malignant lesions. Size <4cm is more likely benign, Hounsfield units <10 usually benign.

BONUS Answers

- 7) B - DI presenting with high volume (>400 cc/h typically for two consecutive hours) dilute (measure with urine osmolality or specific gravity) and hypernatremia. AI is associated with hyponatremia. Hypothyroidism and AI also unlikely to present with symptoms of dilute urine.
- 8) C - primary hyperpara. Hydrochlorothiazide often unmasks underlying PHPT. If it was malignancy or Vit D intoxication, the PTH be low.
- 9) D – You need both glucocorticoid and mineralocorticoid replacement in outpatient management of primary AI. Only at high doses ie hydrocortisone 50-100mg IV, it also has mineracorticoid effect.
- 10) A - Macroprolactin is the next best step: biologically inactive, but measured by assays leading to a false positive. f her prolactin is truly elevated, would then proceed to MRI sella and visual testing especially if symptoms or optic chiasm compression on imaging. Also, usually prolactin is higher in macroadenomas (prolactin 75-100+)
- 11) D - Management of AFF 1) Stop bisphosphonate 2) Treatment with teriparatide for fracture healing, especially if still high risk osteoporosis on BMD. 3) Should also involve ortho assessment especially if complete fracture.

BONUS Answers

12) A - Alendronate is indicated for this patient with new vertebral fracture and clinical risk factors (ongoing smoking). Exercise and resistance training is recommended at least TWICE weekly per Osteoporosis Canada 2023. Vitamin D 1000 IU daily is insufficient to replace the degree of Vit D deficiency this patient has. Denosumab not preferred first line, additionally = High risk of hypocalcemia without correcting Vitamin D.

13) C - Increase glargine to 22 units at bedtime to improve the fasting glucose, and increase aspart to 9 units at breakfast and supper (to target the high sugar pre-lunch and the high at bedtime)

14) D - With history of PAD, his LDLc is at target (<1.8 mmol/L) but his Apo-B is above target (<0.7) and because it is >0.80 , would prefer PCSK9i over Ezetemibe (if pt has \$\$\$ or good drug plan!)

15) 4_ Cinacalcet. -Primary (and Tertiary) hyperparathyroidism can be managed with cinacalcet when patient is not a parathyroidectomy surgical candidate. Calcitriol (activated vit D) is more appropriate in secondary hyperparathyroidism when PTH elevation is due to vitamin D deficiency. IV bisphosphonate may be indicated to improve BMD. In this case her BMD is normal. **Lytic lesions caused by HyperPTH = "Brown tumors"**